WESTERN SYDNEY UNIVERSITY

School of Computer, Data and Mathematical Sciences

COMP 7006 Data Science

Computer Based Assignment – PART A

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QUESTIONS FORMAT:	Word processed document in PDF format; logically presenting answers to each question incorporating R outputs including graphs and charts.
TOTAL MARKS:	**65 Marks (**60 for report + 5 for organisation, presentation of R codes and outputs in Appendix**)**
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TOTAL PAGES:	50

INSTRUCTIONS

Please note that you are expected to answer the questions clearly in this document. Use the template included where relevant to answer. Give the R outputs, comments, and discussion clearly and logically.

Attach all the R commands and outputs in the Appendix.

Write the resulting model equation to the relevant questions with R outputs, especially when you are asked to give the resultant model whenever applicable.

Once completed submit the answer scripts as a **PDF** via the link within vUWS site.

Please note that **5 Marks** is allocated for organization, reasoning, logical flow, and the inclusion of all correct R codes and outputs in the Appendix.

NOTE: Academic misconduct will result in serious consequences. Please refer to the subject outline for more details.

SCENARIO

An insurance company aims to predict the amount of property damage claims resulting from bushfires to ensure they maintain sufficient financial reserves for post-disaster recovery. To achieve this, they have collected comprehensive data related to location, property characteristics, and meteorological conditions, as detailed below. This data will be used to create accurate predictions of property damage claims. Accurate prediction is essential for the company to manage its financial resources effectively, fulfill client expectations, and maintain operational stability in the face of disasters.

Data Description:

Variables	Description
Fire_Intensity	The intensity of the fire measured in kilowatts per metre (kW/m), representing the heat output during the fire event.
Distance_from_Fire	The distance of the building from the fire's origin in kilometres (km)
Building_Age	The number of years since the building was constructed
Property_Value	The estimated value of the property in hundred thousand Australian Dollars ($\times 10^5$ AUD)
Population_Density	The number of people living per square kilometre in the area where the building is located
Emergency_Response_Time	The time taken for emergency services to respond, in minutes

Mitigation_Measures	The number of proactive steps taken to reduce fire risk, such as fireproofing materials, which contribute to minimizing fire damage.
Construction_Quality	A categorical variable assessing the quality of the building, classified as either "Good" or "Bad," based on materials, design, and compliance with safety standards.
Insurance_Coverage	The type of insurance coverage the property owner has, categorized as "Fully," "Partially," or "None," reflecting the level of financial protection available.
Wind_Speed	The average wind speed recorded during the bushfire event in ms^{-1}
Humidity	The average relative humidity during the bushfire in percentage (%)
Temperature	The average temperature in degrees Celsius (^{o}C) during the bushfire event
Damage_Claims	The monetary value of damage claims submitted after the fire, in hundred thousand Australian Dollars (($\times 10^5$AUD), indicating the financial cost of the fire damage.

* Please note that this is a simulated data generated to resemble the real-world data for the purpose of this assignment.

Consider the scenario described and the data set provided [***BushFireData.csv***] to answer the following questions.

1. Identify the target variable and clearly specify the research question. **(3 Marks)**

Target variable: Damage_Claims

Research Question: What are the key factors influencing the monetary value of bushfire damage claims?

2. Understand the data and perform the necessary data pre-processing. Clearly write the steps you have taken to pre-process the given dataset. [Note: make sure to divide the data into training and test set etc.,] **(6 Marks)**

- **Steps taken to pre-process the dataset**

Step 1: Load the dataset using the function:

```
bushfire <- read.csv("BushFireData.csv")
```

Step 2: Check the structure of the dataset to identify data types using str() function.

```
str(bushfire)
```

Step 3: Remove the rows using the function na.omit () which have missing value to get a clean dataset.

```
bushfire <- na.omit(bushfire)
```

Step 4: Convert the categorical variables such as Insurance coverage and construction Quality to factors

```
bushfire$Construction_Quality <- as.factor(bushfire$Construction_Quality)

bushfire$Insurance_Coverage <- as.factor(bushfire$Insurance_Coverage)
```

Step 5: Split the data into training(70%) and testing(30%) ratio with a set.speed().

```
set.seed(1)

tr.id = sample(1:nrow(bushfire), round(nrow(bushfire) * 0.7, 0))

training_data = bushfire[tr.id, ]
testing_data = bushfire[-tr.id, ]
```

Step 6: check the dimensions of the training and testing datasets using the function dim( ) to confirm the split.

```
dim(training_data)

dim(testing_data)S
```

Step 7: Display the structure of the training dataset after pre-processing

```
str(training_data)
```

- **Print the structure of the data before cleaning and pre-processing here**

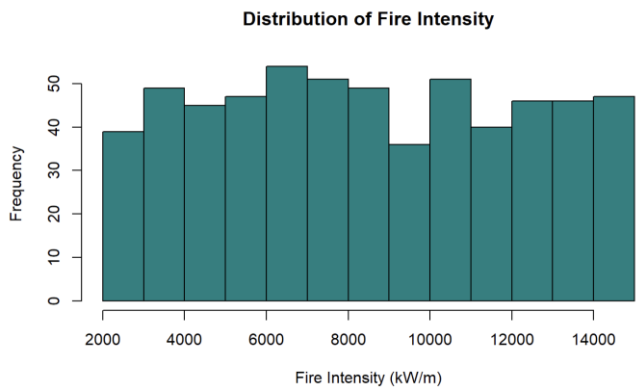
```
str(bushfire)

## 'data.frame':    600 obs. of  13 variables:
##  $ ClaimID                : chr  "DQTA26862" "RDAQ32534" "WTLA41817" "YHZL163
17" ...
##  $ Fire_Intensity         : num  5738 12248 7317 13479 14226 ...
##  $ Distance_from_Fire     : num  5.57 5.49 5.12 4.89 6.13 ...
##  $ Building_Age           : int  19 27 16 30 21 27 24 24 36 30 ...
##  $ Property_Value         : num  9.76 5.89 3.98 4.91 9.31 ...
##  $ Population_Density     : int  520 433 634 576 644 476 500 591 487 441 ...
##  $ Emergency_Response_Time: int  16 18 17 14 12 10 16 19 15 18 ...
##  $ Mitigation_Measures    : int  3 3 4 2 2 1 5 7 2 3 ...
##  $ Construction_Quality   : chr  "Good" "Good" "Good" "Good" ...
##  $ Insurance_Coverage     : chr  "Fully" "None" "Partially" "Partially" ...
##  $ Wind_Speed             : num  24.9 21 25.3 22.3 37.2 31.8 27.8 22.5 32.2 2
0.6 ...
##  $ Humidity               : num  35.3 28.2 44.5 53.8 73.8 ...
##  $ Damage_Claims          : num  7.02 7.45 4.3 7.78 9.46 4.54 7.14 7.73 6.03
5.37 ...
```

- **Print the structure of the training data after cleaning and pre-processing here**

```
str(training_data)

## 'data.frame':    420 obs. of  13 variables:
##  $ ClaimID                : chr  "DBYE20076" "IOFB32666" "DUZA63480" "YKSD931
61" ...
##  $ Fire_Intensity         : num  14819 8696 6165 6006 6399 ...
##  $ Distance_from_Fire     : num  5.54 8.22 5.85 6.94 4.22 5.95 5.12 4.09 7.04
6.6 ...
##  $ Building_Age           : int  31 32 18 30 32 29 16 27 22 31 ...
##  $ Property_Value         : num  8.31 7.55 6.55 5.77 6.54 6.1 3.98 10.1 7.68
5.54 ...
##  $ Population_Density     : int  658 551 586 463 550 646 634 560 730 484 ...
##  $ Emergency_Response_Time: int  3 19 14 23 13 9 17 14 7 18 ...
##  $ Mitigation_Measures    : int  6 2 2 4 6 4 4 2 6 4 ...
##  $ Construction_Quality   : Factor w/ 2 levels "Bad","Good": 2 2 2 1 1 2 2 2
1 2 ...
##  $ Insurance_Coverage     : Factor w/ 3 levels "Fully","None",..: 1 3 1 1 1 1
3 3 3 1 ...
##  $ Wind_Speed             : num  31.3 26.6 27.3 22.8 27.5 18.9 25.3 24.6 14.1
18.6 ...
##  $ Humidity               : num  34.4 67 42.3 46.8 85 ...
##  $ Damage_Claims          : num  5.9 6.59 6.29 4.9 10.95 ...
```

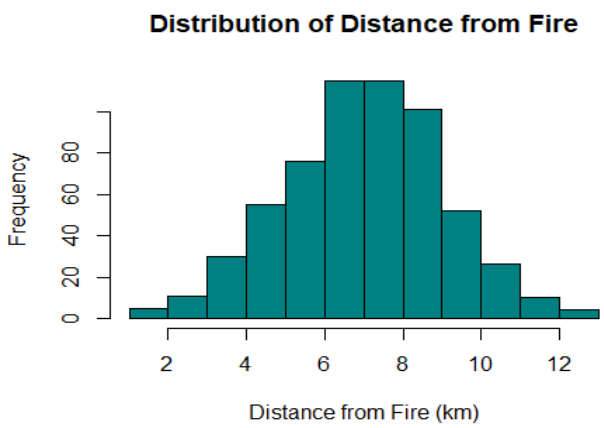
3. Perform a thorough data exploration using the given dataset. You may use various visualization techniques (such as histograms, scatter plots, box plots, correlation matrices, etc.) to uncover interesting patterns and insights. Interpret all your outputs and discuss key findings. [Hint: You may use as many plots as necessary and make sure to interpret them.] **(10 Marks)**

- **Histogram to check the Distribution of Fire_intensity**



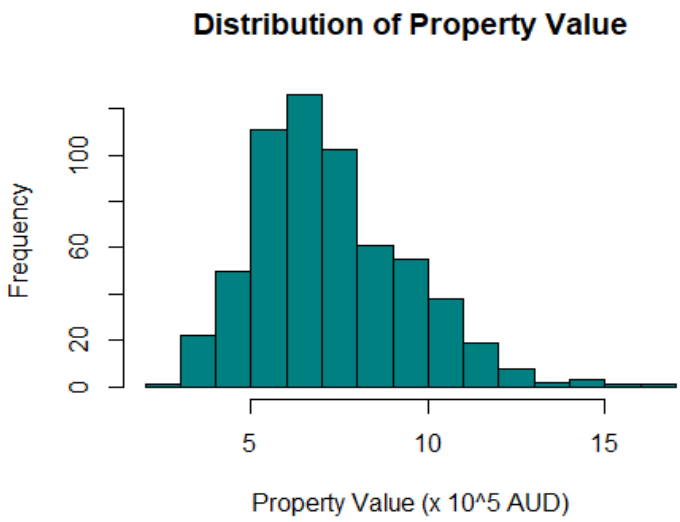
The Histogram of Fire Intensity appears uniform with values ranging from 2000 to close to 15000 kW/m, suggesting that Bushfire intensities are evenly spread across the dataset, indicating no specific skew towards higher or lower values.

- **Histogram to check Distribution of Distance from Fire**



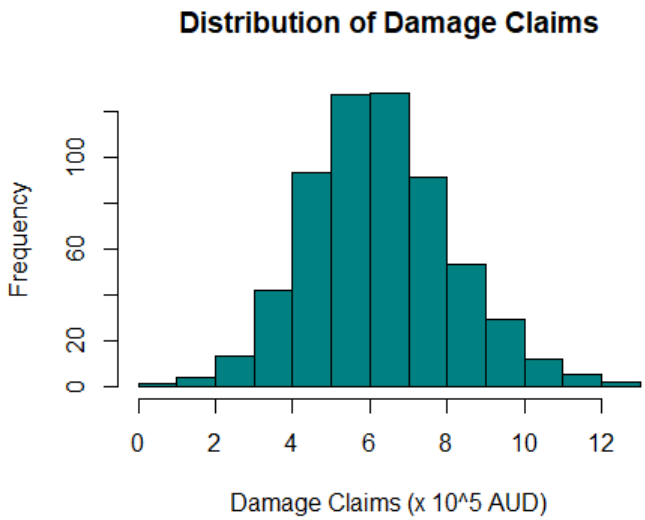
The Histogram of Distance_from_Fire is slightly left-skewed, which means most of the properties are located away from fire, around 6 to 8 kilometers. and fewer properties are located close to fire, which is below 4 kilometers. This indicates that properties nearer to the fire might face higher damages.

- **Histogram to check distribution of property_Value**



The Property_Value histogram is right-skewed, showing that most properties are valued lower, mostly around 5 to 7 (x 10^5 AUD), with fewer high-value properties reaching up to 16. This means we have a lot more lower-valued properties, but high-value ones might have a big impact on damage claims due to their higher worth.

- **Histogram to check distribution of Damage_claims**

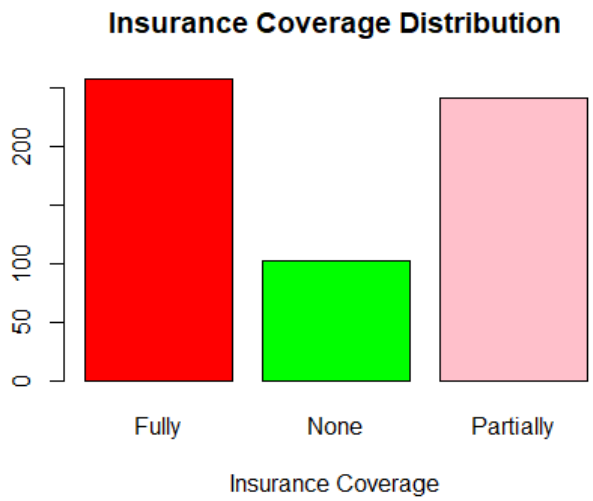


The histogram of damage claims shows a roughly normal distribution centered around 6 (x10^5 AUD), suggesting that most of the claims fall in a moderate range, with less extreme low or high claim values.

- **Bar plot to show distribution of Construction_quality**

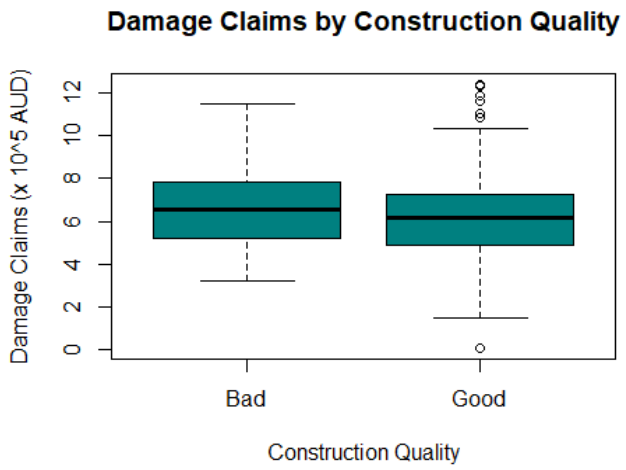
The bar plot shows that most properties are labeled as having "Good" construction quality, with a smaller number listed as "Bad." This imbalance suggests that we may not have enough data on "Bad" quality properties to fully understand how construction quality affects damage claims. Generally, better-quality buildings might be expected to sustain less damage.

- **Barplot to show distribution of Insurance Coverage**



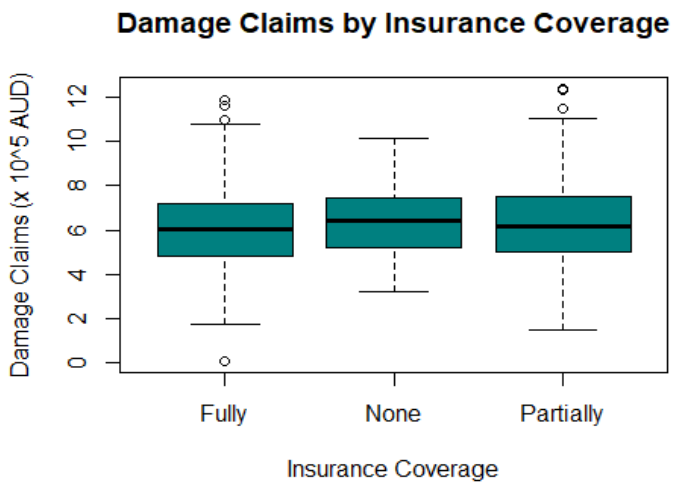
The bar plot for Insurance Coverage shows that most properties are either fully or partially insured, while a smaller portion has no insurance coverage. This is important data, because different levels of insurance might affect the final claim amounts, with fully insured properties possibly having more processed claims.

- **Boxplot to show Damage Claims by Construction Quality**



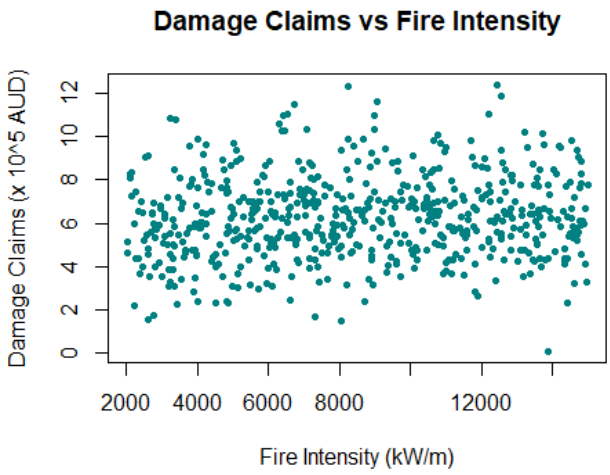
This box plot compares Damage_Claims for "Good" and "Bad" quality buildings. The median claim amounts show the same for both, but "Good" quality buildings have more high-value outliers, which is above 8, which means that while most "Good" buildings have standard claims, some face higher damages. This could mean quality alone doesn't lower damage but may play a role in outlier cases.

- **Boxplot to show Damage claims by Insurance Coverage**



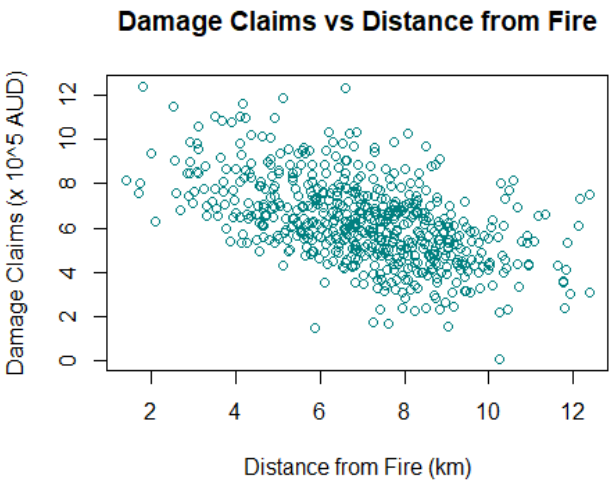
In the box plot the damage claims are approximately similar when compared to 3 of the coverage. However, Fully and partially insured properties have some higher outliers, but on average, there's no major difference in claims due to insurance coverage type. This suggests that while insurance type might not lower or increase the average claims, some insured properties do have higher claims.

- **Scatter plot to check how Fire intensity affects Damage claims**



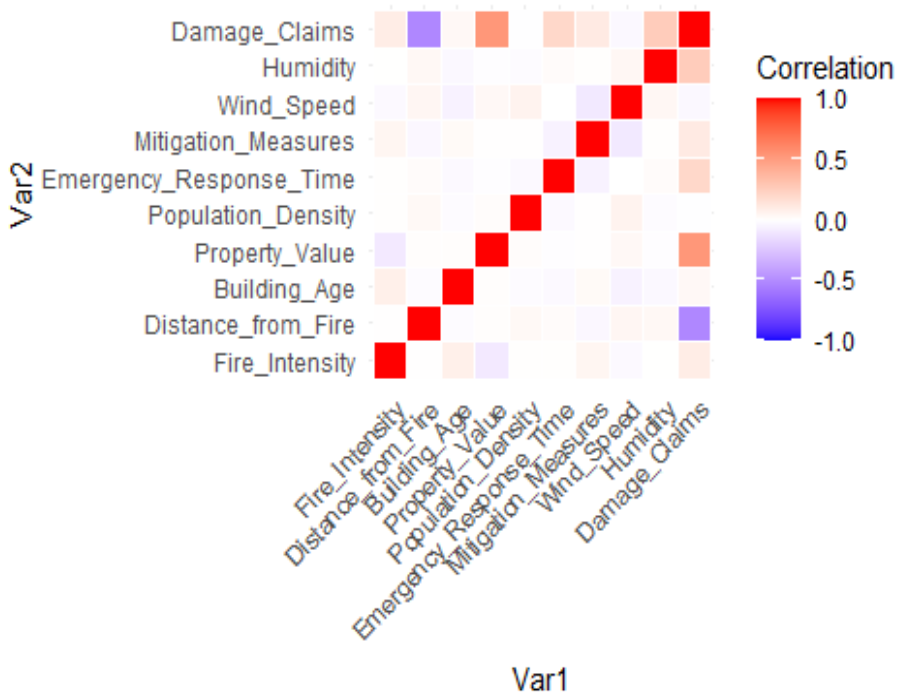
This scatter diagram shows a central scatter, which suggests that majority of properties have Fire_Intensity values between 5,000 and 10,000 kW/m and Damage_Claims between 5 and 8 (×10^5 AUD). This central clustering tells us that for many bushfires, damage claims stay at a moderate level regardless of whether the fire intensity is on the lower or higher side of this range.

- **Scatter plot to check how Distance from Fire affects Damage claims**



The scatter plot shows a downward trend which indicates Properties that are about 2 to 4 kilometers from the fire usually have damage claims over 8 (×10^5 AUD), and some even go up to 10 (×10^5 AUD). This shows that being close to the fire increases the risk of serious damage and higher costs.

- **Correlation matrix to check relationship between variables**



The diagram indicates that there is a moderately strong negative correlation which is around -0.52 between Distance_from_Fire and Damage_Claims. This means that as properties are farther from the fire, their damage claims tend to be lower. The main idea behind that is distance from the fire reduces the risk of serious damage. On the other hand, Property_Value has a positive correlation of about 0.54 with Damage_Claims. This means that higher-value properties tend to have higher damage claims. This could be because high-value properties incur more costly repairs. And, the other variables, like Fire_Intensity, don't show a very strong relationship with Damage_Claims, which confirms that distance and property value are the more important predictors here.

4. Use multiple linear regression to answer the research question. Clearly write the process or all the steps you have taken. Justify your decisions and steps taken whenever necessary [*Hint: model building, model improvement including interaction term and polynomial terms, variable selection, evaluation*]. **(10 Marks)**

Step 1: Build the initial model using all the variables. This will help to understand the relationship between damage claims and other predictors.

```
initial_model <- lm(Damage_Claims ~ Fire_Intensity + Distance_from_Fire +
                      Building_Age + Property_Value + Population_Density +
                      Emergency_Response_Time + Mitigation_Measures +
                      Wind_Speed + Humidity, data=training_data)
```

Step 2: Review the summary of the model using the function Summary (Initial_model) to check each of the predictors (p-values), and check the R-squared and Adjusted R-squared values.

```
summary(initial_model)

##
## Call:
## lm(formula = Damage_Claims ~ Fire_Intensity + Distance_from_Fire +
##     Building_Age + Property_Value + Population_Density + Emergency_Response_T
ime +
##     Mitigation_Measures + Wind_Speed + Humidity, data = training_data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -2.84061 -0.65306  0.01164  0.65415  3.01617
##
## Coefficients:
##                           Estimate Std. Error t value Pr(>|t|)
## (Intercept)              2.138e+00  6.102e-01   3.504 0.000508 ***
## Fire_Intensity           6.746e-05  1.328e-05   5.081 5.71e-07 ***
## Distance_from_Fire      -5.072e-01  2.374e-02 -21.364  < 2e-16 ***
## Building_Age             6.862e-03  9.751e-03   0.704 0.481978
## Property_Value           5.114e-01  2.321e-02  22.032  < 2e-16 ***
## Population_Density       1.938e-04  6.277e-04   0.309 0.757679
## Emergency_Response_Time  9.485e-02  1.221e-02   7.767 6.53e-14 ***
## Mitigation_Measures      8.983e-02  2.803e-02   3.205 0.001458 **
## Wind_Speed              -2.084e-03  9.650e-03  -0.216 0.829097
## Humidity                 2.933e-02  2.612e-03  11.227  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.004 on 410 degrees of freedom
## Multiple R-squared:  0.7308, Adjusted R-squared:  0.7249
## F-statistic: 123.7 on 9 and 410 DF,  p-value: < 2.2e-16
```

- The initial model has an R-squared value of 0.7308, meaning it explains about 73.08% of the variation in Damage_Claims. Significant predictors include Fire_Intensity , Distance_from_Fire, and Property_Value which has p value < 0.05, among others, which means these variables have an impact on Damage_Claims.

Step 3: Model Improvement by adding Interaction Terms

Added an interaction term between Fire_Intensity and Distance_from_Fire to see if their combined effect influences Damage_Claims.

```
interaction_model <- lm(Damage_Claims ~ Fire_Intensity * Distance_from_Fire
+ Building_Age + Property_Value + Population_Density + Emergency_Response_Time +
Mitigation_Measures + Wind_Speed + Humidity, data=training_data)

summary(interaction_model)

##
## Call:
## lm(formula = Damage_Claims ~ Fire_Intensity * Distance_from_Fire +
##     Building_Age + Property_Value + Population_Density + Emergency_Response_T
ime +
##     Mitigation_Measures + Wind_Speed + Humidity, data = training_data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -2.95159 -0.62234  0.01924  0.65670  2.98592
##
## Coefficients:
##                                    Estimate Std. Error t value Pr(>|t|)
## (Intercept)                       1.539e+00  7.415e-01   2.076  0.03855 *
## Fire_Intensity                    1.330e-04  4.805e-05   2.767  0.00591 **
## Distance_from_Fire               -4.290e-01  5.997e-02  -7.154 3.93e-12 ***
## Building_Age                      7.021e-03  9.739e-03   0.721  0.47135
## Property_Value                    5.107e-01  2.319e-02  22.026  < 2e-16 ***
## Population_Density                2.024e-04  6.269e-04   0.323  0.74698
## Emergency_Response_Time           9.460e-02  1.220e-02   7.754 7.12e-14 ***
## Mitigation_Measures               8.979e-02  2.800e-02   3.207  0.00145 **
## Wind_Speed                       -7.954e-04  9.681e-03  -0.082  0.93456
## Humidity                          2.951e-02  2.612e-03  11.298  < 2e-16 ***
## Fire_Intensity:Distance_from_Fire -9.182e-06  6.473e-06  -1.418  0.15681
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.002 on 409 degrees of freedom
## Multiple R-squared:  0.7321, Adjusted R-squared:  0.7256
## F-statistic: 111.8 on 10 and 409 DF,  p-value: < 2.2e-16
```

- The interaction term has a p-value of 0.15681, indicating it's not statistically significant. The R-squared value only slightly increases to 0.7321, so the model doesn't improve much. This suggests that the combined effect of Fire_Intensity and Distance_from_Fire doesn't significantly impact Damage_Claims, so we decide not to keep the interaction term.

Step 4: Adding polynomial model for further improvement

Added a polynomial term for Fire_Intensity to test for a non-linear relationship with Damage_Claims.

```
polynomial_model <- lm(Damage_Claims ~ poly(Fire_Intensity, 2) + Distance_from_F
ire +  Building_Age + Property_Value + Population_Density +
Emergency_Response_Time + Mitigation_Measures + Wind_Speed + Humidity, data=trai
ning_data)

summary(polynomial_model)

##
## Call:
## lm(formula = Damage_Claims ~ poly(Fire_Intensity, 2) + Distance_from_Fire +
##     Building_Age + Property_Value + Population_Density + Emergency_Response_T
ime +
##     Mitigation_Measures + Wind_Speed + Humidity, data = training_data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -2.75271 -0.66278  0.01094  0.63759  3.01351
##
## Coefficients:
##                            Estimate Std. Error t value Pr(>|t|)
## (Intercept)               2.7354882  0.6012813   4.549 7.10e-06 ***
## poly(Fire_Intensity, 2)1  5.1251520  1.0096593   5.076 5.86e-07 ***
## poly(Fire_Intensity, 2)2 -0.9547977  1.0230059  -0.933  0.35120
## Distance_from_Fire       -0.5066524  0.0237512 -21.332  < 2e-16 ***
## Building_Age              0.0073168  0.0097642   0.749  0.45408
## Property_Value            0.5101282  0.0232546  21.937  < 2e-16 ***
## Population_Density        0.0001732  0.0006282   0.276  0.78293
## Emergency_Response_Time   0.0932225  0.0123397   7.555 2.77e-13 ***
## Mitigation_Measures       0.0922713  0.0281585   3.277  0.00114 **
## Wind_Speed               -0.0017565  0.0096581  -0.182  0.85578
## Humidity                  0.0293251  0.0026125  11.225  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.004 on 409 degrees of freedom
## Multiple R-squared:  0.7314, Adjusted R-squared:  0.7248
## F-statistic: 111.4 on 10 and 409 DF,  p-value: < 2.2e-16
```

- The first-degree polynomial term is significant, but the second-degree term has a p-value of 0.3512, which is not significant. The R-squared of 0.7314 shows minimal improvement. This suggests that a non-linear effect of Fire_Intensity does not add much value, so we choose to exclude the polynomial term.

Step 5: Using stepwise regression to refine the model and for selecting significant predictors.

```
library(MASS)
optimized_model <- stepAIC(initial_model, direction="both")

## Start:  AIC=13.01
## Damage_Claims ~ Fire_Intensity + Distance_from_Fire + Building_Age +
##     Property_Value + Population_Density + Emergency_Response_Time +
##     Mitigation_Measures + Wind_Speed + Humidity
##
##                            Df Sum of Sq    RSS    AIC
## - Wind_Speed                1      0.05 413.11  11.05
## - Population_Density        1      0.10 413.16  11.10
## - Building_Age              1      0.50 413.56  11.51
## <none>                                413.06  13.01
## - Mitigation_Measures       1     10.35 423.41  21.40
## - Fire_Intensity            1     26.01 439.08  36.66
## - Emergency_Response_Time   1     60.77 473.83  68.65
## - Humidity                  1    127.00 540.06 123.60
## - Distance_from_Fire        1    459.82 872.88 325.25
## - Property_Value            1    489.04 902.11 339.08
##
## Step:  AIC=11.05
## Damage_Claims ~ Fire_Intensity + Distance_from_Fire + Building_Age +
##     Property_Value + Population_Density + Emergency_Response_Time +
##     Mitigation_Measures + Humidity
##
##                            Df Sum of Sq    RSS    AIC
## - Population_Density        1      0.09 413.20   9.14
## - Building_Age              1      0.51 413.62   9.57
## <none>                                413.11  11.05
## + Wind_Speed                1      0.05 413.06  13.01
## - Mitigation_Measures       1     10.58 423.69  19.67
## - Fire_Intensity            1     26.21 439.32  34.89
## - Emergency_Response_Time   1     60.72 473.83  66.65
## - Humidity                  1    127.16 540.28 121.77
## - Distance_from_Fire        1    459.92 873.03 323.32
## - Property_Value            1    489.00 902.11 337.08
##
## Step:  AIC=9.14
## Damage_Claims ~ Fire_Intensity + Distance_from_Fire + Building_Age +
##     Property_Value + Emergency_Response_Time + Mitigation_Measures +
##     Humidity
##
##                            Df Sum of Sq    RSS    AIC
## - Building_Age              1      0.52 413.72   7.67
## <none>                                413.20   9.14
## + Population_Density        1      0.09 413.11  11.05
## + Wind_Speed                1      0.04 413.16  11.10
## - Mitigation_Measures       1     10.58 423.78  17.76
## - Fire_Intensity            1     26.32 439.52  33.08
## - Emergency_Response_Time   1     60.68 473.88  64.70
## - Humidity                  1    127.10 540.30 119.78
## - Distance_from_Fire        1    460.82 874.02 321.80
## - Property_Value            1    488.91 902.11 335.08
##
## Step:  AIC=7.67
## Damage_Claims ~ Fire_Intensity + Distance_from_Fire + Property_Value +
```

```
##     Emergency_Response_Time + Mitigation_Measures + Humidity
##
##                            Df Sum of Sq    RSS    AIC
## <none>                                  413.72   7.67
## + Building_Age              1      0.52 413.20   9.14
## + Population_Density        1      0.10 413.62   9.57
## + Wind_Speed                1      0.05 413.67   9.62
## - Mitigation_Measures       1     10.55 424.27  16.25
## - Fire_Intensity            1     26.87 440.58  32.10
## - Emergency_Response_Time   1     60.31 474.03  62.83
## - Humidity                  1    127.09 540.81 118.18
## - Distance_from_Fire        1    461.88 875.60 320.56
## - Property_Value            1    489.23 902.95 333.47
```

```r
summary(optimized_model)
```

```
##
## Call:
## lm(formula = Damage_Claims ~ Fire_Intensity + Distance_from_Fire +
##     Property_Value + Emergency_Response_Time + Mitigation_Measures +
##     Humidity, data = training_data)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -2.8343 -0.6784  0.0130  0.6447  3.0003
##
## Coefficients:
##                            Estimate Std. Error t value Pr(>|t|)
## (Intercept)               2.374e+00  3.580e-01   6.629 1.06e-10 ***
## Fire_Intensity            6.831e-05  1.319e-05   5.179 3.50e-07 ***
## Distance_from_Fire       -5.072e-01  2.362e-02 -21.473  < 2e-16 ***
## Property_Value            5.114e-01  2.314e-02  22.099  < 2e-16 ***
## Emergency_Response_Time   9.438e-02  1.216e-02   7.759 6.76e-14 ***
## Mitigation_Measures       9.030e-02  2.783e-02   3.245  0.00127 **
## Humidity                  2.928e-02  2.600e-03  11.264  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.001 on 413 degrees of freedom
## Multiple R-squared:  0.7304, Adjusted R-squared:  0.7265
## F-statistic: 186.5 on 6 and 413 DF,  p-value: < 2.2e-16
```

- The optimized model includes Fire_Intensity, Distance_from_Fire, Property_Value, Emergency_Response_Time, Mitigation_Measures, and Humidity. The R-squared value is 0.7304, almost the same as the initial model, but with fewer predictors. This streamlined model is more efficient and still effective at predicting Damage_Claims.

Step 6: Calculated the Root Mean Square Error (RMSE) on the test data to assess the model's accuracy.

```
predictions <- predict(optimized_model, testing_data)
rmse <- sqrt(mean((testing_data$Damage_Claims - predictions)^2))
rmse

## [1] 1.002394
```

- The RMSE is 1.002, which means that, on average, the model's predictions of Damage_Claims deviate by around 1.002 units (in hundreds of thousands AUD). This indicates a reasonably good prediction accuracy for the optimized model.

5. Give your resultant model. **(3 Marks)**

```
final_model <- lm(Damage_Claims ~ Fire_Intensity + Distance_from_Fire +
Property_Value + Emergency_Response_Time + Mitigation_Measures + Humidit
y, data = training_data)

summary(final_model)

##
## Call:
## lm(formula = Damage_Claims ~ Fire_Intensity + Distance_from_Fire +
##     Property_Value + Emergency_Response_Time + Mitigation_Measures +
##     Humidity, data = training_data)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -2.8343 -0.6784  0.0130  0.6447  3.0003
##
## Coefficients:
##                           Estimate Std. Error t value Pr(>|t|)
## (Intercept)              2.374e+00  3.580e-01   6.629 1.06e-10 ***
## Fire_Intensity           6.831e-05  1.319e-05   5.179 3.50e-07 ***
## Distance_from_Fire      -5.072e-01  2.362e-02 -21.473  < 2e-16 ***
## Property_Value           5.114e-01  2.314e-02  22.099  < 2e-16 ***
## Emergency_Response_Time  9.438e-02  1.216e-02   7.759 6.76e-14 ***
## Mitigation_Measures      9.030e-02  2.783e-02   3.245  0.00127 **
## Humidity                 2.928e-02  2.600e-03  11.264  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.001 on 413 degrees of freedom
## Multiple R-squared:  0.7304, Adjusted R-squared:  0.7265
## F-statistic: 186.5 on 6 and 413 DF,  p-value: < 2.2e-16
```

```r
predictions <- predict(final_model, testing_data)
final_rmse <- sqrt(mean((testing_data$Damage_Claims - predictions)^2))
cat("Final Model RMSE:", final_rmse, "\n")
```

```
## Final Model RMSE: 1.002394
```

- The final model for predicting Damage_Claims includes the predictors: Fire_Intensity, Distance_from_Fire, Property_Value, Emergency_Response_Time, Mitigation_Measures, and Humidity. This model achieves an R-squared value of approximately 0.7304, meaning it explains around 73% of the variance in Damage_Claims, and it has a low Root Mean Square Error (RMSE) of about 1.0024 on the test data. These values indicate that the model provides reliable predictions with a strong fit between the predictors and the target variable.

6. A researcher in the company is curious in exploring the factors influencing the damage claims. Discuss your results based on your findings. (**5 Marks**)

Factors influencing the damage claim are identified on the basis of findings from Multiple linear regression model. It identified the key predictors that impact the monetary value of bushfire damage claims. Following are the factors influencing damage_claims:

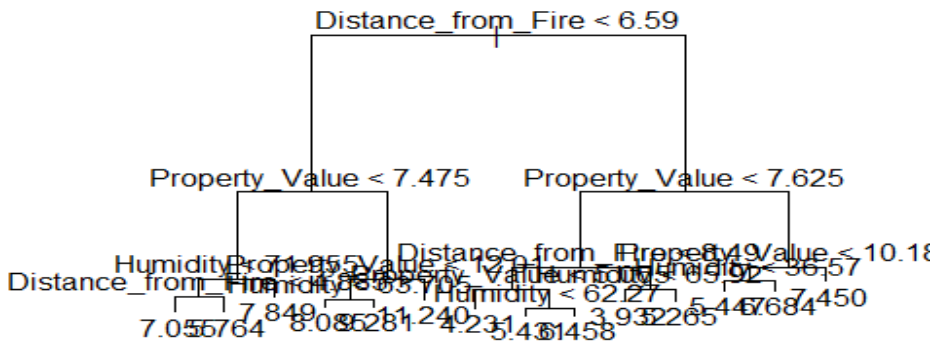
- Fire Intensity: An increase in fire intensity results in higher damage claims.
- Distance from Fire: Properties closer to the fire tend to have higher claims, as being nearer increases the risk of severe damage.
- Property Value: Higher-value properties have larger claims, likely because these properties are more costly to repair or replace.
- Emergency Response Time: Longer response times are associated with higher claims, as delays allow the fire more time to cause extensive damage.
- Mitigation Measures: Properties with more fire prevention measures show a slight increase in claims, possibly because higher-risk properties are more likely to implement these measures.
- Humidity: Humidity has a smaller effect, potentially influencing fire behavior and indirectly affecting claim amounts.

7. Use decision tree to answer the research question. Clearly outline and explain each step of the process involved [*Hint: model building, improvement and evaluation*]. **(8 Marks)**

Step 1: Building the Initial Decision Tree Model

Created an initial decision tree model with Damage_Claims as the target variable and all other available predictors and plotted the model

```
tree_train <- tree(Damage_Claims~.,training_data)

plot(tree_train)
text(tree_train, pretty = 0)
```




```
summary(tree_train)

##
## Regression tree:
## tree(formula = Damage_Claims ~ ., data = training_data)
## Variables actually used in tree construction:
## [1] "Distance_from_Fire" "Property_Value"     "Humidity"
## Number of terminal nodes:  14
## Residual mean deviance:  1.352 = 549 / 406
## Distribution of residuals:
##     Min.  1st Qu.   Median     Mean  3rd Qu.     Max.
## -4.26400 -0.80600  0.02716  0.00000  0.76590  3.19900
```

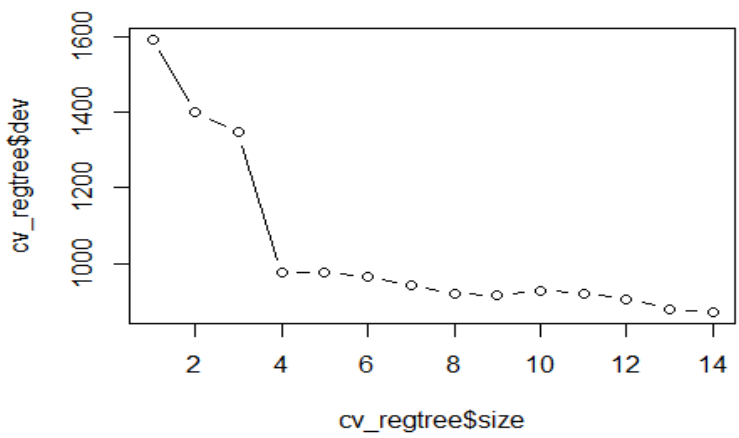
- The plot of the decision tree provides a visual representation of how the model splits the data based on the most important predictors for Damage_Claims. We can see that initial model has 14 terminal nodes. This is relatively high in number. The tree firstly splits on Distance_from_Fire, followed by Property_Value and Humidity. This indicates that these variables are key factors in predicting Damage_Claims, as the tree selects them repeatedly to divide the data. And the Residual Mean Deviance is around

1352 , which reflects the average variance of prediction errors. A lower deviance will be the ideal one, as it indicates better fit.

Step 2: Improving the Model Using Cross-Validation

To refine the model and prevent overfitting, we performed cross-validation. This helps us determine the ideal number of terminal nodes by evaluating different model complexities and their performance

```
set.seed(12)
cv_regtree <- cv.tree(tree_train)

cv_regtree

## $size
##  [1] 14 13 12 11 10  9  8  7  6  5  4  3  2  1
##
## $dev
##  [1]  871.4151  878.2538  905.4496  921.3159  928.3191  915.3571  921.3180
##  [8]  941.5276  964.9441  976.3285  976.3436 1346.1607 1399.4211 1591.6324
##
## $k
##  [1]      -Inf  15.82732  16.33324  20.42914  26.43445  29.27510  32.80486
##  [8]  36.91487  39.83928  41.93083  46.04175 160.37850 185.23915 334.14506
##
## $method
## [1] "deviance"
##
## attr(,"class")
## [1] "prune"         "tree.sequence"

plot(cv_regtree$size, cv_regtree$dev, type = "b" )
```



- From the plot, we see that deviance initially decreases as the tree size grows but begins to follow the same level around 9 terminal nodes. This suggests that a 9-node tree

minimizes the deviance without unnecessary complexity. Based on the cross-validation results, we select 9 nodes as the optimal tree size for simplicity and accuracy.

Step 3: Used the findings from cross-validation, we pruned the tree to 9 nodes.

```
# Prune to size 15 for minimal deviance
pruned_tree_9 <- prune.tree(tree_train, best = 9)
predictions_9 <- predict(pruned_tree_9, testing_data)
```

#Check the model Accuracy

```
MSE_9 <- mean((testing_data$Damage_Claims - predictions_9)^2)
RMSE_9 <- sqrt(MSE_9)
RMSE_9
```

```
## [1] 1.461267
```

- The RMSE score finds a good fit for our regression tree model, so lets proceed with the model with size 9.

Summarise the best model Regression Tree

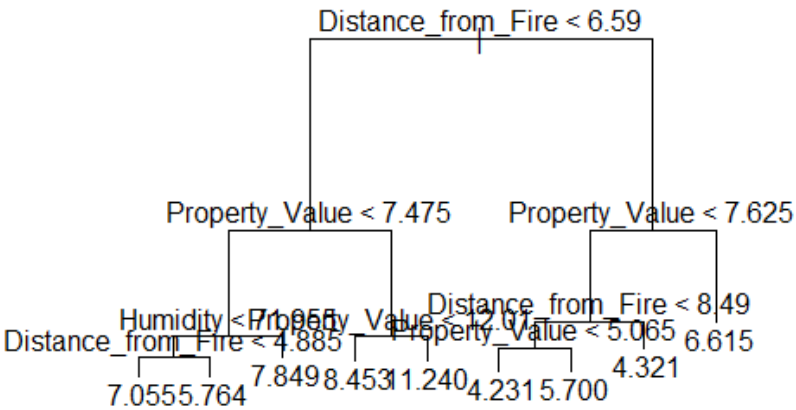
```
pruned_tree_9

## node), split, n, deviance, yval
##       * denotes terminal node
##
##  1) root 420 1535.000  6.178
##    2) Distance_from_Fire < 6.59 160  530.300  7.315
##      4) Property_Value < 7.475 102  213.200  6.504
##        8) Humidity < 71.955 85  139.200  6.235
##         16) Distance_from_Fire < 4.885 31   33.590  7.055 *
##         17) Distance_from_Fire > 4.885 54   72.790  5.764 *
##        9) Humidity > 71.955 17   37.090  7.849 *
##      5) Property_Value > 7.475 58  131.800  8.742
##       10) Property_Value < 12.01 52   81.490  8.453 *
##       11) Property_Value > 12.01 6    8.417 11.240 *
##    3) Distance_from_Fire > 6.59 260  670.200  5.478
##      6) Property_Value < 7.625 176  359.600  4.936
##       12) Distance_from_Fire < 8.49 104  205.900  5.361
##         24) Property_Value < 5.065 24   33.210  4.231 *
##         25) Property_Value > 5.065 80  132.900  5.700 *
##       13) Distance_from_Fire > 8.49 72  107.600  4.321 *
##      7) Property_Value > 7.625 84  150.200  6.615 *
```

- Starting at the root, the first split at Distance_from_Fire of 6.59 km indicates that properties closer to the fire generally incur higher claims. For properties within this distance, additional splits based on Property_Value and Humidity provide further detail. For instance, properties with lower Property_Value and Humidity under 71.955 tend to have average claims around 6.235, while higher humidity levels increase this

average to 7.849. Conversely, properties farther than 6.59 km tend to have lower claims, with further splits by Property_Value. Notably, properties with very high Property_Value (over 12.01) exhibit the highest average claims at 11.24. Overall, the model highlights Distance_from_Fire, Property_Value, and Humidity as critical determinants of Damage_Claims.

Step 4: regression tree model is plotted here

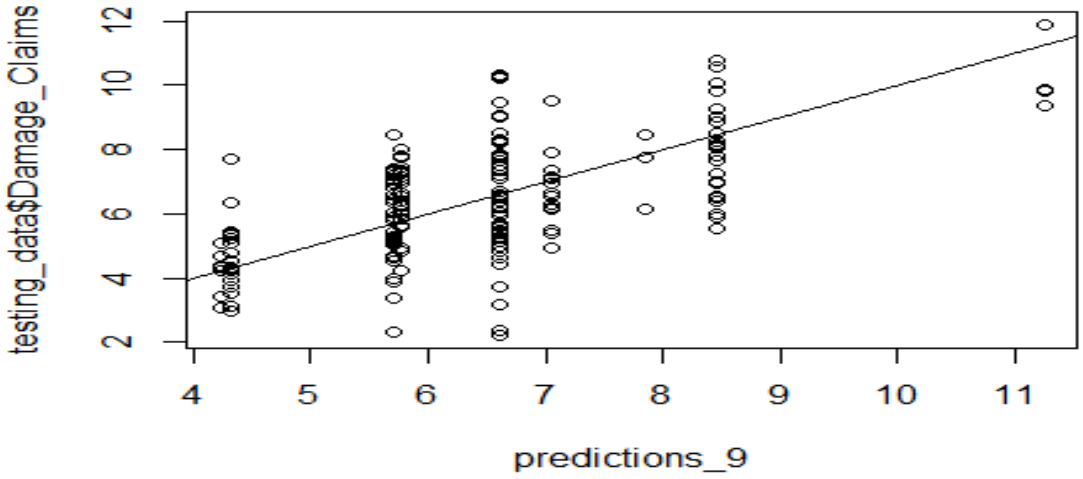
```
plot(pruned_tree_9)
text(pruned_tree_9, pretty = 0)
```



- Model shows that Distance_from_Fire, Property_Value, and Humidity are the primary factors affecting Damage_Claims. The tree first split at the Distance_from_Fire < 6.59, indicating that properties near to the fire which tend to have higher claims, with further splits based on property value and humidity to refine predictions.

Step 5: plotted the prediction

```
plot(predictions_9, testing_data$Damage_Claims)
abline(0, 1)
```

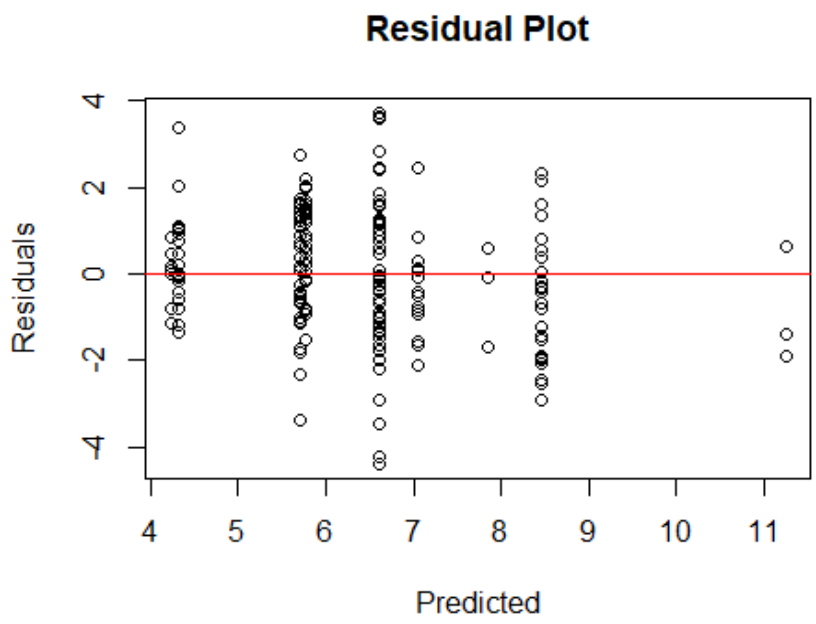


- This plot compares predicted and actual Damage_Claims, where closer alignment to the reference line indicates better accuracy. While predictions generally follow the trend, the spread around values (e.g., 6 and 7) suggests some variation in accuracy, particularly for mid-range claims.

Step 6: we checked whether the model is biased.

Residuals should be randomly scattered around zero; patterns may suggest areas where the model struggles.

```
residuals <- testing_data$Damage_Claims - predictions_9
plot(predictions_9,
     residuals, xlab = "Predicted", ylab = "Residuals", main = "Residual Plot")
abline(h = 0, col = "red")
```



- This residual plot shows the difference between actual and predicted values, with residuals centered around zero. Ideally, residuals should be randomly scattered without a clear pattern, which indicates that the model captures the data trends well. While there is some spread in the residuals, particularly around certain predicted values like 6 and 9, the plot overall suggests that the model's errors are not biased, as residuals are evenly distributed above and below the red line.

8. Give the resultant model and interpret it. Clearly describe the terminal nodes. **(5 Marks)**

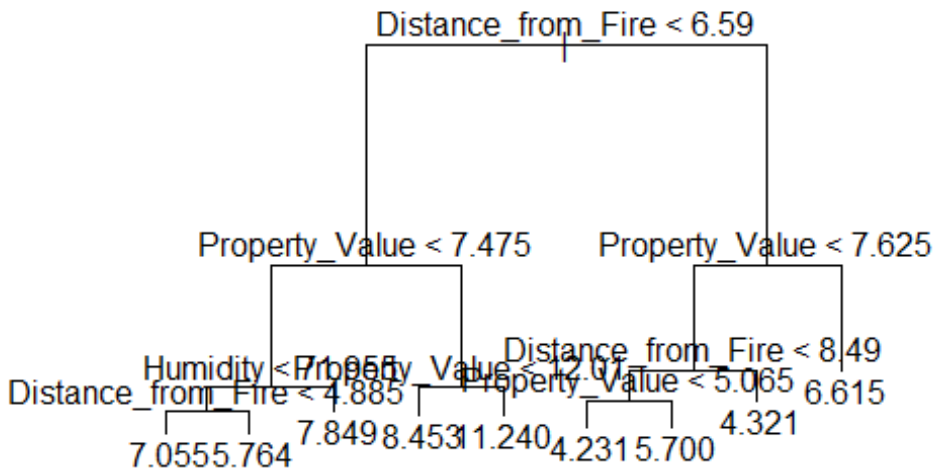
- The pruned decision tree model, with 9 terminal nodes, reveals that `Distance_from_Fire`, `Property_Value`, and `Humidity` are key factors influencing `Damage_Claims`, with closer proximity to fire, higher property values, and lower humidity generally leading to higher claims. Following are the terminal nodes and resultant tree model.

```
#Terminal nodes

pruned_tree_9

## node), split, n, deviance, yval
##       * denotes terminal node
##
##  1) root 420 1535.000  6.178
##    2) Distance_from_Fire < 6.59 160  530.300  7.315
##      4) Property_Value < 7.475 102  213.200  6.504
```

```
##         8) Humidity < 71.955 85  139.200  6.235
##          16) Distance_from_Fire < 4.885 31   33.590  7.055 *
##          17) Distance_from_Fire > 4.885 54   72.790  5.764 *
##         9) Humidity > 71.955 17   37.090  7.849 *
##       5) Property_Value > 7.475 58  131.800  8.742
##        10) Property_Value < 12.01 52   81.490  8.453 *
##        11) Property_Value > 12.01 6    8.417 11.240 *
##     3) Distance_from_Fire > 6.59 260  670.200  5.478
##       6) Property_Value < 7.625 176  359.600  4.936
##        12) Distance_from_Fire < 8.49 104  205.900  5.361
##          24) Property_Value < 5.065 24   33.210  4.231 *
##          25) Property_Value > 5.065 80  132.900  5.700 *
##        13) Distance_from_Fire > 8.49 72  107.600  4.321 *
##       7) Property_Value > 7.625 84  150.200  6.615 *
```

Interpretting the model with size 9

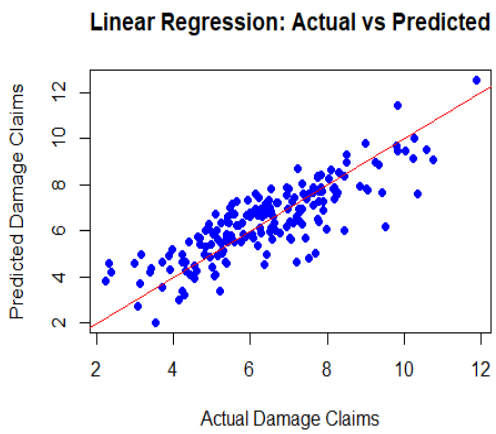
```
plot(pruned_tree_9)
text(pruned_tree_9, pretty = 0)
```

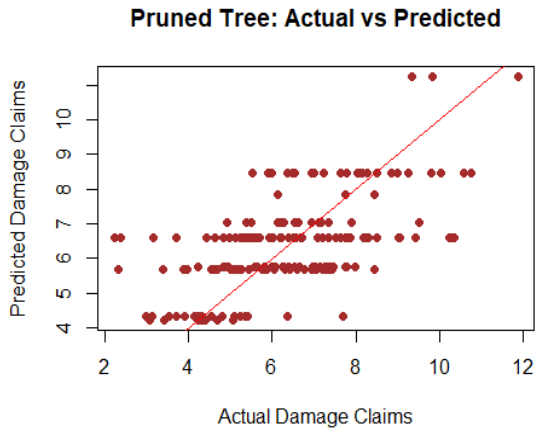


9. Compare the performance of different resultant models (Question 5 and 8) you obtained above. Use at least 1 performance measure and a graphical representation. **(5 Marks)**

Model	RMSE	Prediction Accuracy	Graphical Representation

Linear Regression Model	1.002	Higher accuracy due to lower RMSE, indicating closer predictions to actual values	Predicted vs. Actual plot shows points close to the 45-degree line, demonstrating a good fit
Pruned Regression Tree Model	1.461	Lower accuracy with a higher RMSE, indicating greater prediction error	Predicted vs. Actual plot shows a wider spread of points, demonstrating less precise predictions





10. The company is now interested in correctly classifying whether the damage claim would be greater than 7.5 hundred thousand AUD or not. Modify the target variable to address this scenario. Hence pre-process the data accordingly. Give a suitable technique that can be used to address this scenario. **(5 Marks)**

Print the structure of the modified dataset. Remove redundant or unnecessary variables and include the new target variable. [Hint: use *str()* function and give the R output here]

```{r}
str(bushfire)
```

```
'data.frame':   600 obs. of  14 variables:
 $ ClaimID                : chr  "DQTA26862" "RDAQ32534" "WTLA41817" "YHZL16317" ...
 $ Fire_Intensity         : num  5738 12248 7317 13479 14226 ...
 $ Distance_from_Fire     : num  5.57 5.49 5.12 4.89 6.13 ...
 $ Building_Age           : int  19 27 16 30 21 27 24 24 36 30 ...
 $ Property_Value         : num  9.76 5.89 3.98 4.91 9.31 ...
 $ Population_Density     : int  520 433 634 576 644 476 500 591 487 441 ...
 $ Emergency_Response_Time: int  16 18 17 14 12 10 16 19 15 18 ...
 $ Mitigation_Measures    : int  3 3 4 2 2 1 5 7 2 3 ...
 $ Construction_Quality   : Factor w/ 2 levels "Bad","Good": 2 2 2 2 2 2 2 2 2 2 ...
 $ Insurance_Coverage     : Factor w/ 3 levels "Fully","None",..: 1 2 3 3 2 1 3 1 1 3 ...
 $ Wind_Speed             : num  24.9 21 25.3 22.3 37.2 31.8 27.8 22.5 32.2 20.6 ...
 $ Humidity               : num  35.3 28.2 44.5 53.8 73.8 ...
 $ Damage_Claims          : num  7.02 7.45 4.3 7.78 9.46 4.54 7.14 7.73 6.03 5.37 ...
 $ NewDamageClaims        : num  0 0 0 1 1 0 0 1 0 0 ...
```

Our new variable name NewDamageClaims consist of binary values 0 and 1, When the threshold 7.5 hundred thousand in Damage claims less than 7.5 its 0 and when greater its 1.

New Target Variable: NewDamageClaims.

Suitable technique: Logistic Regression can be a best fit or else A decision tree can be a choice.

--- End of questions for Part A. Part B will be available soon ---

APPENDIX

```r
library(melt)
library(reshape2)
library(psych)
library(ggplot2)

library(tree)
```

Question 2: Data Pre-processing Steps

Load the dataset

```r
bushfire <- read.csv("BushFireData.csv")
```

Check the structure of the dataset to identify data types

```r
str(bushfire)

## 'data.frame':    600 obs. of  13 variables:
##  $ ClaimID                : chr  "DQTA26862" "RDAQ32534" "WTLA41817" "YHZL16317
"  ...
##  $ Fire_Intensity         : num  5738 12248 7317 13479 14226 ...
##  $ Distance_from_Fire     : num  5.57 5.49 5.12 4.89 6.13 ...
##  $ Building_Age           : int  19 27 16 30 21 27 24 24 36 30 ...
##  $ Property_Value         : num  9.76 5.89 3.98 4.91 9.31 ...
##  $ Population_Density     : int  520 433 634 576 644 476 500 591 487 441 ...
```

```
##  $ Emergency_Response_Time: int  16 18 17 14 12 10 16 19 15 18 ...
##  $ Mitigation_Measures    : int  3 3 4 2 2 1 5 7 2 3 ...
##  $ Construction_Quality   : chr  "Good" "Good" "Good" "Good" ...
##  $ Insurance_Coverage     : chr  "Fully" "None" "Partially" "Partially" ...
##  $ Wind_Speed             : num  24.9 21 25.3 22.3 37.2 31.8 27.8 22.5 32.2 20.
6 ...
##  $ Humidity               : num  35.3 28.2 44.5 53.8 73.8 ...
##  $ Damage_Claims          : num  7.02 7.45 4.3 7.78 9.46 4.54 7.14 7.73 6.03 5.
37 ...
```

Remove rows with missing values to ensure a clean dataset for analysis

```
bushfire <- na.omit(bushfire)
```

Convert categorical variables (Construction Quality and Insurance Coverage) to factors

```
bushfire$Construction_Quality <- as.factor(bushfire$Construction_Quality)

bushfire$Insurance_Coverage <- as.factor(bushfire$Insurance_Coverage)
```

Set a seed for reproducibility and create a 70-30 train-test split

```
set.seed(1)

tr.id = sample(1:nrow(bushfire), round(nrow(bushfire) * 0.7, 0))
```

Define the training data (70% of the dataset) and testing data (remaining 30%)

```
training_data = bushfire[tr.id, ]  # Training data
testing_data = bushfire[-tr.id, ]  # Test data
```

Check the dimensions of the training and testing datasets to confirm the split

```
dim(training_data)

## [1] 420  13

dim(testing_data)

## [1] 180  13
```

Display the structure of the training dataset after pre-processing

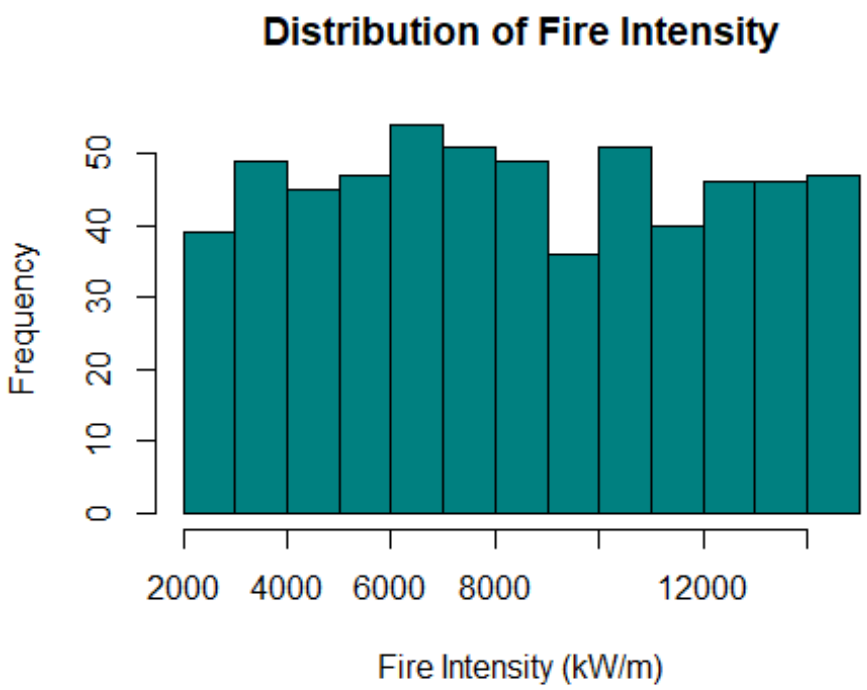
```
str(training_data)

## 'data.frame':    420 obs. of  13 variables:
##  $ ClaimID                : chr  "TEUQ67875" "XRIJ46118" "UTCV67354" "BSLD89476
" ...
##  $ Fire_Intensity         : num  3845 14285 3221 3288 8536 ...
##  $ Distance_from_Fire     : num  10.27 7.55 3.66 9.36 7.16 ...
##  $ Building_Age           : int  32 36 21 23 21 27 24 30 29 19 ...
##  $ Property_Value         : num  12.52 8.08 10.51 10.16 6.8 ...
##  $ Population_Density     : int  583 670 582 458 560 516 616 699 668 614 ...
##  $ Emergency_Response_Time: int  23 17 14 18 16 12 15 12 13 20 ...
##  $ Mitigation_Measures    : int  1 2 3 1 2 4 4 6 3 3 ...
##  $ Construction_Quality   : Factor w/ 2 levels "Bad","Good": 2 2 2 2 2 2 2 2 1
2 ...
##  $ Insurance_Coverage     : Factor w/ 3 levels "Fully","None",..: 3 1 3 1 3 1 3
1 3 1 ...
##  $ Wind_Speed             : num  23.9 17.8 24.6 19.4 19.6 18.4 26.7 22 21.8 26.
```

```
4 ...
##  $ Humidity                : num  70 48.4 75.5 24.4 64.1 ...
##  $ Damage_Claims           : num  8.04 6.57 10.85 4.38 4.95 ...
```

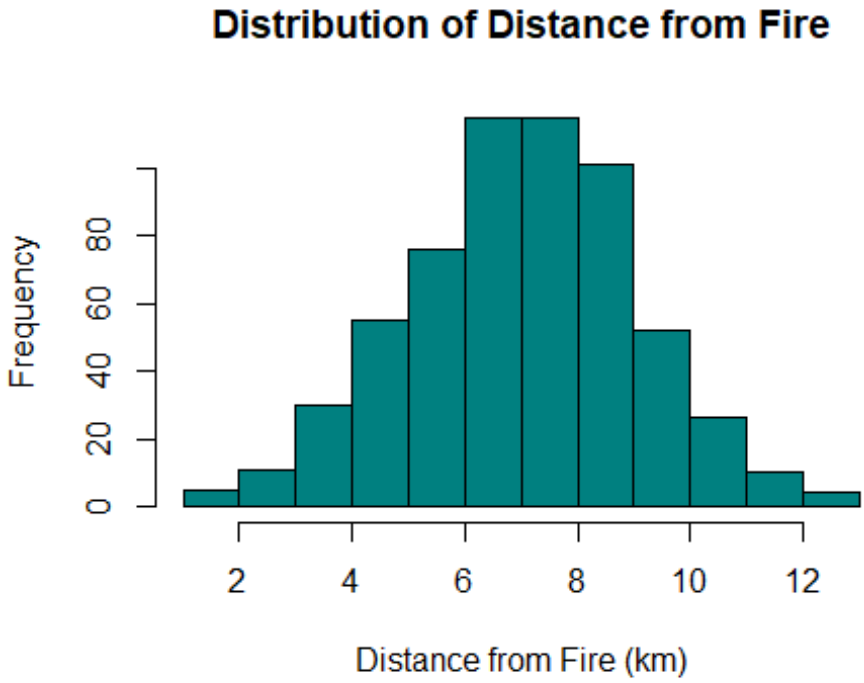
QUESTION 3

#Distribution of Fire Intensity

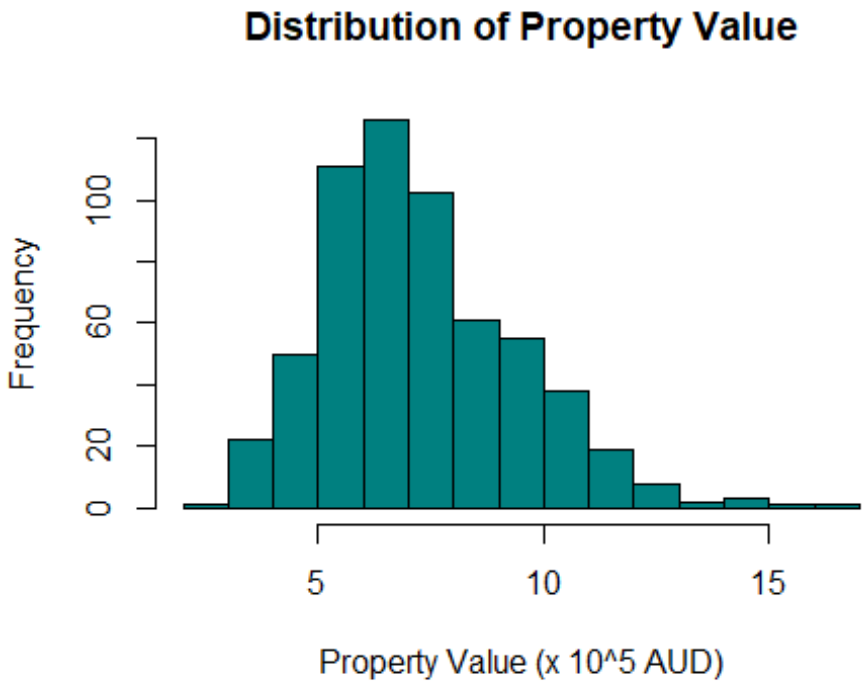
```
hist(bushfire$Fire_Intensity, main = "Distribution of Fire Intensity", xlab = "Fir
e Intensity (kW/m)", col = "#008080")
```



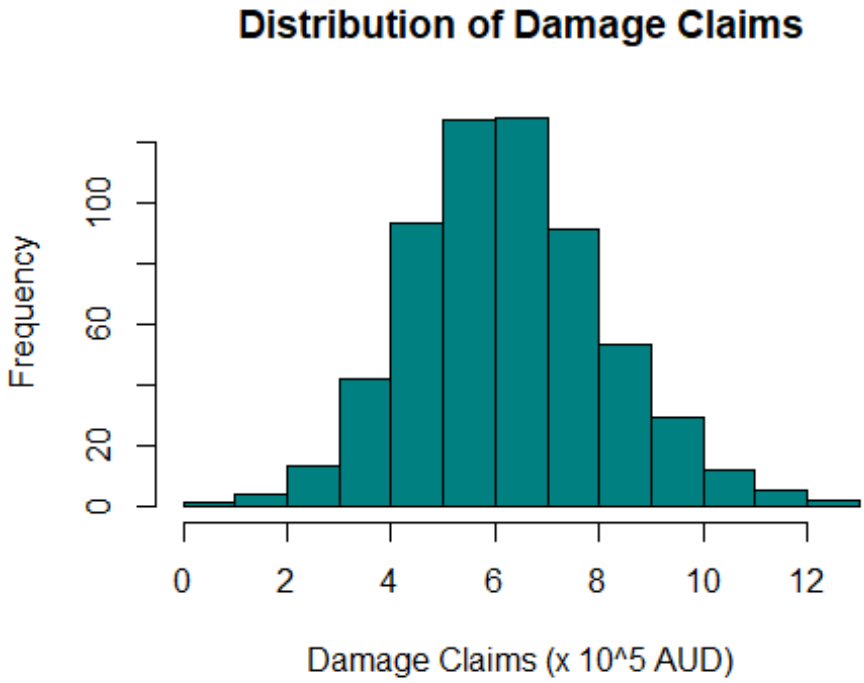

```
hist(bushfire$Distance_from_Fire, main = "Distribution of Distance from Fire", xla
b = "Distance from Fire (km)", col = "#008080")
```




```
hist(bushfire$Property_Value, main = "Distribution of Property Value", xlab = "Pro
perty Value (x 10^5 AUD)", col = "#008080")
```



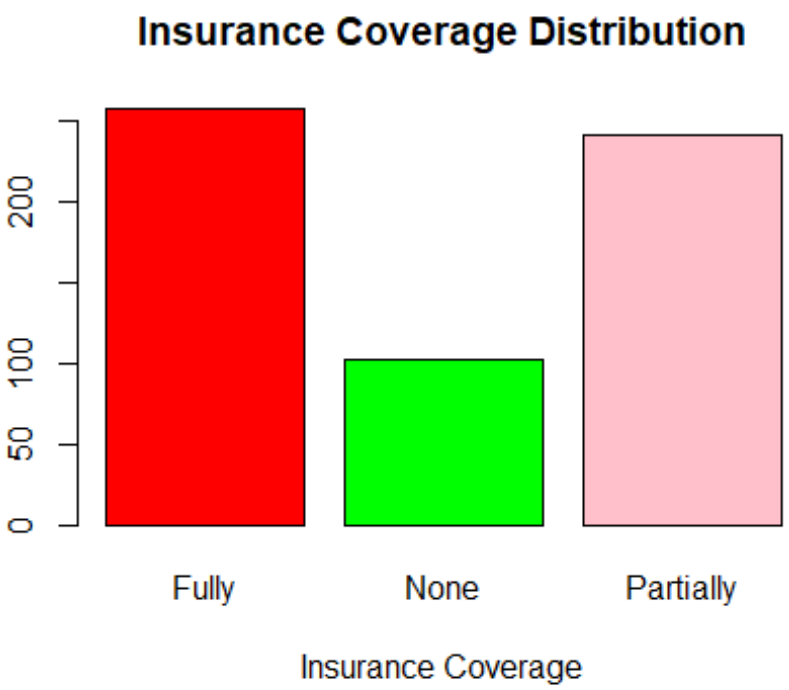

```
hist(bushfire$Damage_Claims, main = "Distribution of Damage Claims", xlab = "Damag
e Claims (x 10^5 AUD)", col = "#008080")
```



#2 bar plots

```
barplot(table(bushfire$Construction_Quality), main = "Construction Quality Distrib
ution", xlab = "Construction Quality", col = c("red", "green"))
```

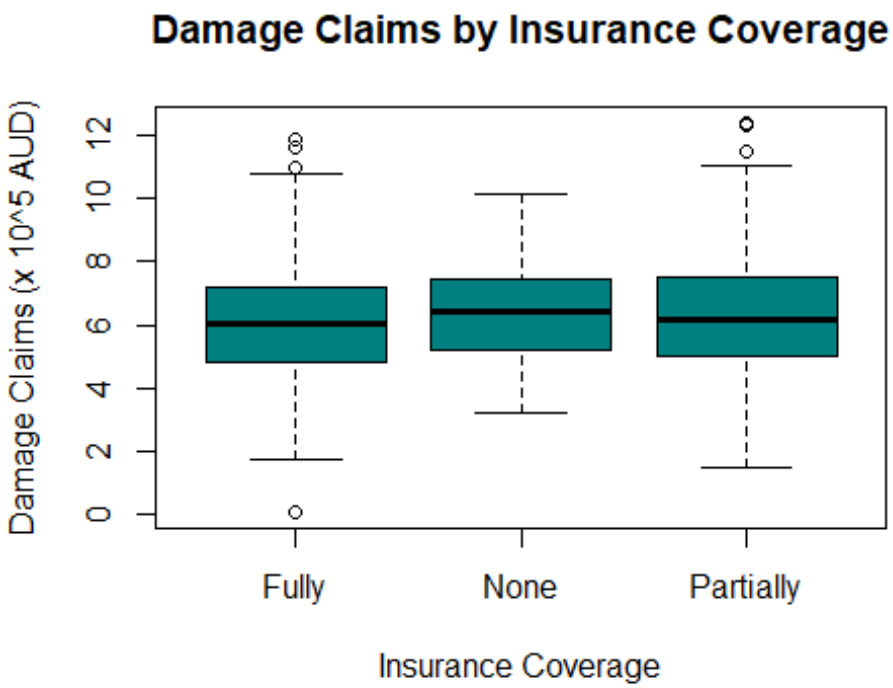
```
barplot(table(bushfire$Insurance_Coverage), main = "Insurance Coverage Distributio
n", xlab = "Insurance Coverage", col = c("red", "green", "pink"))
```



boxplots

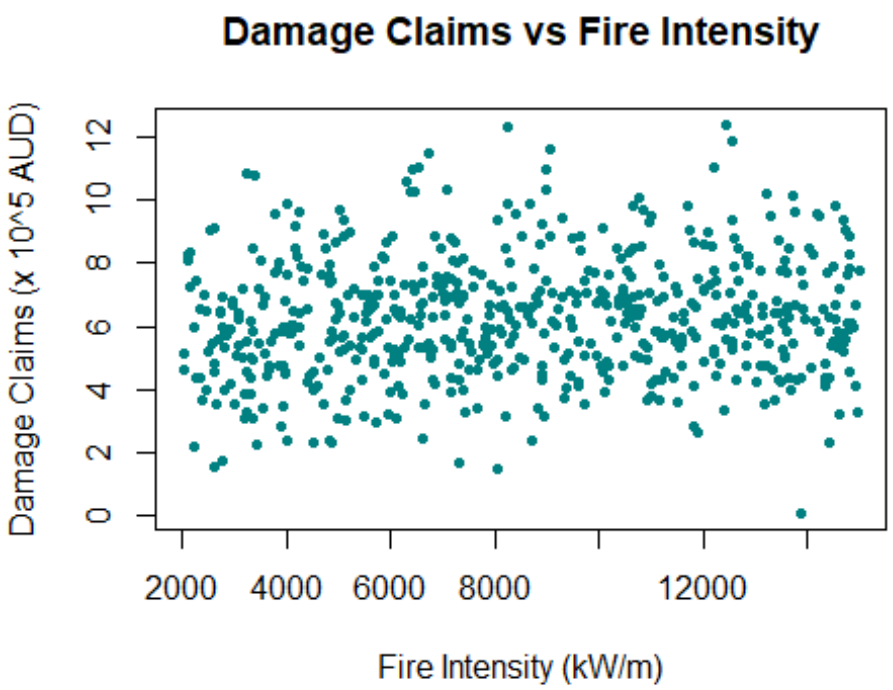
```
boxplot(Damage_Claims ~ Construction_Quality, data = bushfire,
        main = "Damage Claims by Construction Quality",
        xlab = "Construction Quality", ylab = "Damage Claims (x 10^5 AUD)", col =
"#008080")
```

```r
boxplot(Damage_Claims ~ Insurance_Coverage, data = bushfire,
        main = "Damage Claims by Insurance Coverage",
        xlab = "Insurance Coverage", ylab = "Damage Claims (x 10^5 AUD)", col = "#
008080")
```

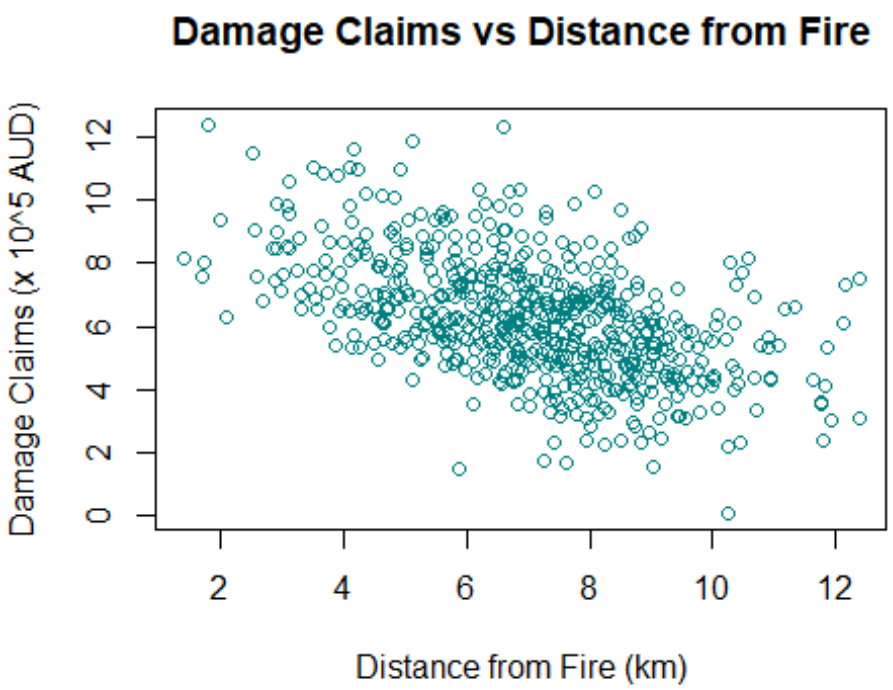


Scatter Plots

```r
plot(bushfire$Fire_Intensity, bushfire$Damage_Claims,
     main = "Damage Claims vs Fire Intensity",
     xlab = "Fire Intensity (kW/m)", ylab = "Damage Claims (x 10^5 AUD)", pch = 20
, col = "#008080")
```




```
plot(bushfire$Distance_from_Fire, bushfire$Damage_Claims,
     main = "Damage Claims vs Distance from Fire",
     xlab = "Distance from Fire (km)", ylab = "Damage Claims (x 10^5 AUD)", col =
"#008080" )
```

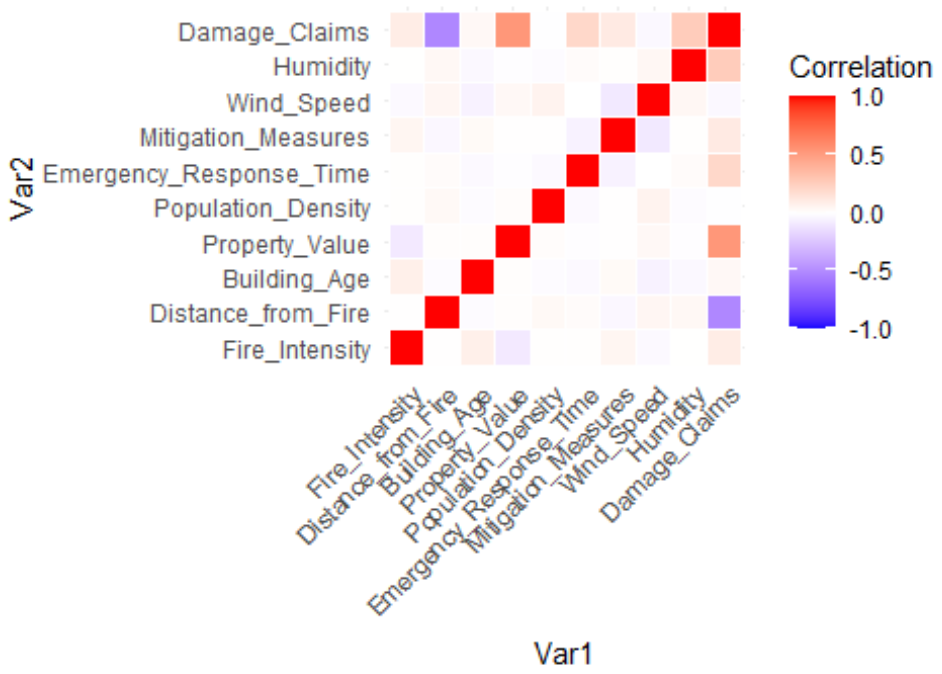



```
# Select only continuous variables
continuous_vars <- bushfire[, sapply(bushfire, is.numeric)]
```

```r
cor_matrix <- cor(continuous_vars, use = "complete.obs")
print(cor_matrix)
```

```
##                          Fire_Intensity Distance_from_Fire Building_Age
## Fire_Intensity              1.000000000         0.00430711   0.07932316
## Distance_from_Fire          0.004307110         1.00000000  -0.01747107
## Building_Age                0.079323163        -0.01747107   1.00000000
## Property_Value             -0.092373683         0.01046487   0.01011021
## Population_Density          0.005966024         0.03368348  -0.01500278
## Emergency_Response_Time     0.006990306         0.01909041  -0.02546560
## Mitigation_Measures         0.048600046        -0.03202978   0.02876564
## Wind_Speed                 -0.026135802         0.04742770  -0.05554963
## Humidity                    0.002108686         0.03929606  -0.02762756
## Damage_Claims               0.094694540        -0.51972220   0.03508000
##                          Property_Value Population_Density
## Fire_Intensity             -0.092373683        0.005966024
## Distance_from_Fire          0.010464875        0.033683480
## Building_Age                0.010110213       -0.015002777
## Property_Value              1.000000000        0.013515053
## Population_Density          0.013515053        1.000000000
## Emergency_Response_Time    -0.007032240       -0.026580657
## Mitigation_Measures         0.005550135        0.004063548
## Wind_Speed                  0.035542394        0.061671042
## Humidity                   -0.008731118       -0.016382216
## Damage_Claims               0.535456227       -0.005300330
##                          Emergency_Response_Time Mitigation_Measures
## Fire_Intensity                       0.006990306         0.048600046
## Distance_from_Fire                   0.019090406        -0.032029785
## Building_Age                        -0.025465605         0.028765635
## Property_Value                      -0.007032240         0.005550135
## Population_Density                  -0.026580657         0.004063548
## Emergency_Response_Time              1.000000000        -0.056227459
## Mitigation_Measures                 -0.056227459         1.000000000
## Wind_Speed                          -0.000547797        -0.093437039
## Humidity                             0.019014280         0.007378899
## Damage_Claims                        0.204320762         0.108466773
##                           Wind_Speed     Humidity Damage_Claims
## Fire_Intensity          -0.026135802  0.002108686    0.09469454
## Distance_from_Fire       0.047427705  0.039296064   -0.51972220
## Building_Age            -0.055549633 -0.027627558    0.03508000
## Property_Value           0.035542394 -0.008731118    0.53545623
## Population_Density       0.061671042 -0.016382216   -0.00530033
## Emergency_Response_Time -0.000547797  0.019014280    0.20432076
## Mitigation_Measures     -0.093437039  0.007378899    0.10846677
## Wind_Speed               1.000000000  0.042964276   -0.02932162
## Humidity                 0.042964276  1.000000000    0.26419933
## Damage_Claims           -0.029321620  0.264199332    1.00000000
```

```r
cor_matrix <- cor(continuous_vars, use = "complete.obs")
melted_cor_matrix <- melt(cor_matrix)

ggplot(data = melted_cor_matrix, aes(x = Var1, y = Var2, fill = value)) +
  geom_tile(color = "white") +
  scale_fill_gradient2(low = "blue", high = "red", mid = "white",
                       midpoint = 0, limit = c(-1, 1), space = "Lab",
                       name = "Correlation") +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, vjust = 1, hjust = 1)) +
  coord_fixed()
```



QUESTION 4

#Initial Model Building

```
initial_model <- lm(Damage_Claims ~ Fire_Intensity + Distance_from_Fire +
                      Building_Age + Property_Value + Population_Density +
                      Emergency_Response_Time + Mitigation_Measures +
                      Wind_Speed + Humidity, data=training_data)

summary(initial_model)

##
## Call:
## lm(formula = Damage_Claims ~ Fire_Intensity + Distance_from_Fire +
##     Building_Age + Property_Value + Population_Density + Emergency_Response_Tim
e +
##     Mitigation_Measures + Wind_Speed + Humidity, data = training_data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -2.84061 -0.65306  0.01164  0.65415  3.01617
##
## Coefficients:
##                           Estimate Std. Error t value Pr(>|t|)
## (Intercept)              2.138e+00  6.102e-01   3.504 0.000508 ***
## Fire_Intensity           6.746e-05  1.328e-05   5.081 5.71e-07 ***
## Distance_from_Fire      -5.072e-01  2.374e-02 -21.364  < 2e-16 ***
## Building_Age             6.862e-03  9.751e-03   0.704 0.481978
## Property_Value           5.114e-01  2.321e-02  22.032  < 2e-16 ***
## Population_Density       1.938e-04  6.277e-04   0.309 0.757679
## Emergency_Response_Time  9.485e-02  1.221e-02   7.767 6.53e-14 ***
## Mitigation_Measures      8.983e-02  2.803e-02   3.205 0.001458 **
## Wind_Speed              -2.084e-03  9.650e-03  -0.216 0.829097
```

```
## Humidity                 2.933e-02  2.612e-03  11.227  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.004 on 410 degrees of freedom
## Multiple R-squared:  0.7308, Adjusted R-squared:  0.7249
## F-statistic: 123.7 on 9 and 410 DF,  p-value: < 2.2e-16
```

Model Improvement

```
interaction_model <- lm(Damage_Claims ~ Fire_Intensity * Distance_from_Fire
+ Building_Age + Property_Value + Population_Density + Emergency_Response_Time + M
itigation_Measures + Wind_Speed + Humidity, data=training_data)

summary(interaction_model)
```

```
##
## Call:
## lm(formula = Damage_Claims ~ Fire_Intensity * Distance_from_Fire +
##     Building_Age + Property_Value + Population_Density + Emergency_Response_Tim
e +
##     Mitigation_Measures + Wind_Speed + Humidity, data = training_data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -2.95159 -0.62234  0.01924  0.65670  2.98592
##
## Coefficients:
##                                     Estimate Std. Error t value Pr(>|t|)
## (Intercept)                        1.539e+00  7.415e-01   2.076  0.03855 *
## Fire_Intensity                     1.330e-04  4.805e-05   2.767  0.00591 **
## Distance_from_Fire                -4.290e-01  5.997e-02  -7.154 3.93e-12 ***
## Building_Age                       7.021e-03  9.739e-03   0.721  0.47135
## Property_Value                     5.107e-01  2.319e-02  22.026  < 2e-16 ***
## Population_Density                 2.024e-04  6.269e-04   0.323  0.74698
## Emergency_Response_Time            9.460e-02  1.220e-02   7.754 7.12e-14 ***
## Mitigation_Measures                8.979e-02  2.800e-02   3.207  0.00145 **
## Wind_Speed                        -7.954e-04  9.681e-03  -0.082  0.93456
## Humidity                           2.951e-02  2.612e-03  11.298  < 2e-16 ***
## Fire_Intensity:Distance_from_Fire -9.182e-06  6.473e-06  -1.418  0.15681
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.002 on 409 degrees of freedom
## Multiple R-squared:  0.7321, Adjusted R-squared:  0.7256
## F-statistic: 111.8 on 10 and 409 DF,  p-value: < 2.2e-16
```

Polynomial model

```
polynomial_model <- lm(Damage_Claims ~ poly(Fire_Intensity, 2) + Distance_from_Fir
e +  Building_Age + Property_Value + Population_Density +
Emergency_Response_Time + Mitigation_Measures + Wind_Speed + Humidity, data=traini
ng_data)

summary(polynomial_model)
```

```
##
## Call:
## lm(formula = Damage_Claims ~ poly(Fire_Intensity, 2) + Distance_from_Fire +
##     Building_Age + Property_Value + Population_Density + Emergency_Response_Tim
```

```
e +
##     Mitigation_Measures + Wind_Speed + Humidity, data = training_data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -2.75271 -0.66278  0.01094  0.63759  3.01351
##
## Coefficients:
##                            Estimate Std. Error t value Pr(>|t|)
## (Intercept)               2.7354882  0.6012813   4.549 7.10e-06 ***
## poly(Fire_Intensity, 2)1  5.1251520  1.0096593   5.076 5.86e-07 ***
## poly(Fire_Intensity, 2)2 -0.9547977  1.0230059  -0.933  0.35120
## Distance_from_Fire       -0.5066524  0.0237512 -21.332  < 2e-16 ***
## Building_Age              0.0073168  0.0097642   0.749  0.45408
## Property_Value            0.5101282  0.0232546  21.937  < 2e-16 ***
## Population_Density        0.0001732  0.0006282   0.276  0.78293
## Emergency_Response_Time   0.0932225  0.0123397   7.555 2.77e-13 ***
## Mitigation_Measures       0.0922713  0.0281585   3.277  0.00114 **
## Wind_Speed               -0.0017565  0.0096581  -0.182  0.85578
## Humidity                  0.0293251  0.0026125  11.225  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.004 on 409 degrees of freedom
## Multiple R-squared:  0.7314, Adjusted R-squared:  0.7248
## F-statistic: 111.4 on 10 and 409 DF,  p-value: < 2.2e-16
```

Variable Selection Using Stepwise Regression

```r
library(MASS)
optimized_model <- stepAIC(initial_model, direction="both")
```

```
## Start:  AIC=13.01
## Damage_Claims ~ Fire_Intensity + Distance_from_Fire + Building_Age +
##     Property_Value + Population_Density + Emergency_Response_Time +
##     Mitigation_Measures + Wind_Speed + Humidity
##
##                           Df Sum of Sq    RSS    AIC
## - Wind_Speed               1      0.05 413.11  11.05
## - Population_Density       1      0.10 413.16  11.10
## - Building_Age             1      0.50 413.56  11.51
## <none>                                413.06  13.01
## - Mitigation_Measures      1     10.35 423.41  21.40
## - Fire_Intensity           1     26.01 439.08  36.66
## - Emergency_Response_Time  1     60.77 473.83  68.65
## - Humidity                 1    127.00 540.06 123.60
## - Distance_from_Fire       1    459.82 872.88 325.25
## - Property_Value           1    489.04 902.11 339.08
##
## Step:  AIC=11.05
## Damage_Claims ~ Fire_Intensity + Distance_from_Fire + Building_Age +
##     Property_Value + Population_Density + Emergency_Response_Time +
##     Mitigation_Measures + Humidity
##
##                           Df Sum of Sq    RSS    AIC
## - Population_Density       1      0.09 413.20   9.14
## - Building_Age             1      0.51 413.62   9.57
## <none>                                413.11  11.05
## + Wind_Speed               1      0.05 413.06  13.01
```

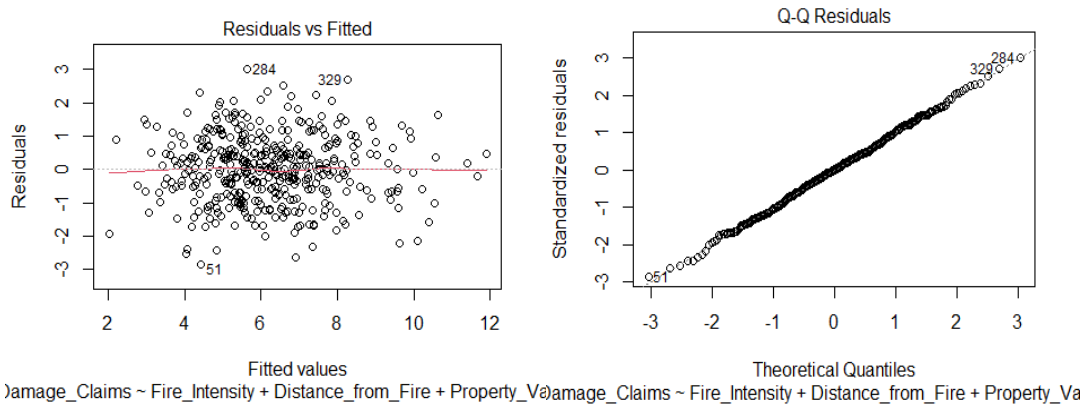
```
## - Mitigation_Measures      1     10.58 423.69  19.67
## - Fire_Intensity           1     26.21 439.32  34.89
## - Emergency_Response_Time  1     60.72 473.83  66.65
## - Humidity                 1    127.16 540.28 121.77
## - Distance_from_Fire       1    459.92 873.03 323.32
## - Property_Value           1    489.00 902.11 337.08
##
## Step:  AIC=9.14
## Damage_Claims ~ Fire_Intensity + Distance_from_Fire + Building_Age +
##     Property_Value + Emergency_Response_Time + Mitigation_Measures +
##     Humidity
##
##                           Df Sum of Sq    RSS    AIC
## - Building_Age             1      0.52 413.72   7.67
## <none>                                 413.20   9.14
## + Population_Density       1      0.09 413.11  11.05
## + Wind_Speed               1      0.04 413.16  11.10
## - Mitigation_Measures      1     10.58 423.78  17.76
## - Fire_Intensity           1     26.32 439.52  33.08
## - Emergency_Response_Time  1     60.68 473.88  64.70
## - Humidity                 1    127.10 540.30 119.78
## - Distance_from_Fire       1    460.82 874.02 321.80
## - Property_Value           1    488.91 902.11 335.08
##
## Step:  AIC=7.67
## Damage_Claims ~ Fire_Intensity + Distance_from_Fire + Property_Value +
##     Emergency_Response_Time + Mitigation_Measures + Humidity
##
##                           Df Sum of Sq    RSS    AIC
## <none>                                 413.72   7.67
## + Building_Age             1      0.52 413.20   9.14
## + Population_Density       1      0.10 413.62   9.57
## + Wind_Speed               1      0.05 413.67   9.62
## - Mitigation_Measures      1     10.55 424.27  16.25
## - Fire_Intensity           1     26.87 440.58  32.10
## - Emergency_Response_Time  1     60.31 474.03  62.83
## - Humidity                 1    127.09 540.81 118.18
## - Distance_from_Fire       1    461.88 875.60 320.56
## - Property_Value           1    489.23 902.95 333.47
```

```r
summary(optimized_model)
```

```
##
## Call:
## lm(formula = Damage_Claims ~ Fire_Intensity + Distance_from_Fire +
##     Property_Value + Emergency_Response_Time + Mitigation_Measures +
##     Humidity, data = training_data)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -2.8343 -0.6784  0.0130  0.6447  3.0003
##
## Coefficients:
##                           Estimate Std. Error t value Pr(>|t|)
## (Intercept)              2.374e+00  3.580e-01   6.629 1.06e-10 ***
## Fire_Intensity           6.831e-05  1.319e-05   5.179 3.50e-07 ***
## Distance_from_Fire      -5.072e-01  2.362e-02 -21.473  < 2e-16 ***
## Property_Value           5.114e-01  2.314e-02  22.099  < 2e-16 ***
## Emergency_Response_Time  9.438e-02  1.216e-02   7.759 6.76e-14 ***
```

```
## Mitigation_Measures      9.030e-02  2.783e-02   3.245  0.00127 **
## Humidity                 2.928e-02  2.600e-03  11.264  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.001 on 413 degrees of freedom
## Multiple R-squared:  0.7304, Adjusted R-squared:  0.7265
## F-statistic: 186.5 on 6 and 413 DF,  p-value: < 2.2e-16
```

```
plot(optimized_model)
```



Model Evaluation with RMSE

```
predictions <- predict(optimized_model, testing_data)
rmse <- sqrt(mean((testing_data$Damage_Claims - predictions)^2))
rmse

## [1] 1.002394
```

QUESTION 5

```
final_model <- lm(Damage_Claims ~ Fire_Intensity + Distance_from_Fire + Property_V
alue + Emergency_Response_Time + Mitigation_Measures + Humidity, data = training_d
ata)

summary(final_model)

##
## Call:
```

```
## lm(formula = Damage_Claims ~ Fire_Intensity + Distance_from_Fire +
##     Property_Value + Emergency_Response_Time + Mitigation_Measures +
##     Humidity, data = training_data)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -2.8343 -0.6784  0.0130  0.6447  3.0003
##
## Coefficients:
##                           Estimate Std. Error t value Pr(>|t|)
## (Intercept)              2.374e+00  3.580e-01   6.629 1.06e-10 ***
## Fire_Intensity           6.831e-05  1.319e-05   5.179 3.50e-07 ***
## Distance_from_Fire      -5.072e-01  2.362e-02 -21.473  < 2e-16 ***
## Property_Value           5.114e-01  2.314e-02  22.099  < 2e-16 ***
## Emergency_Response_Time  9.438e-02  1.216e-02   7.759 6.76e-14 ***
## Mitigation_Measures      9.030e-02  2.783e-02   3.245  0.00127 **
## Humidity                 2.928e-02  2.600e-03  11.264  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.001 on 413 degrees of freedom
## Multiple R-squared:  0.7304, Adjusted R-squared:  0.7265
## F-statistic: 186.5 on 6 and 413 DF,  p-value: < 2.2e-16
```

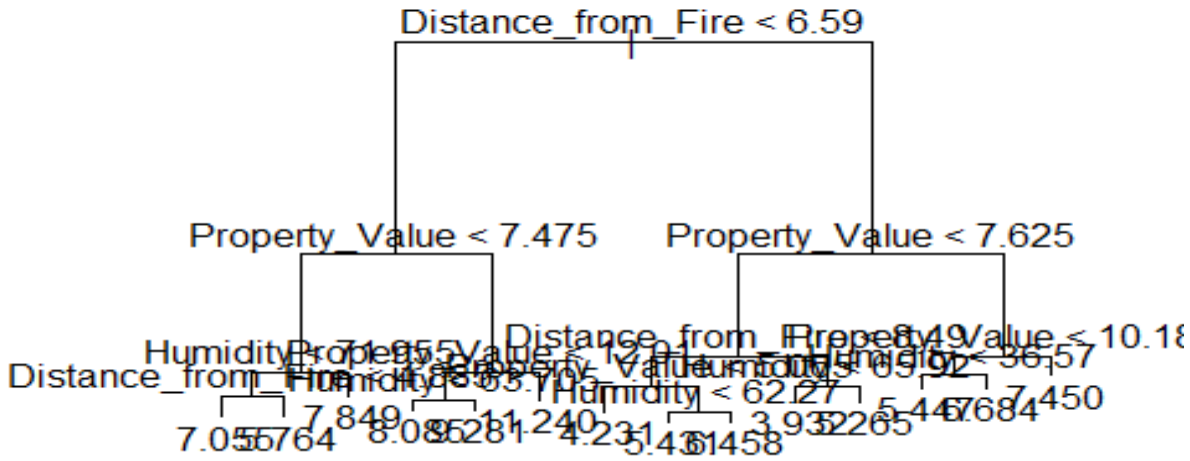
```r
predictions <- predict(final_model, testing_data)
final_rmse <- sqrt(mean((testing_data$Damage_Claims - predictions)^2))
cat("Final Model RMSE:", final_rmse, "\n")
```

```
## Final Model RMSE: 1.002394
```

QUESTION 7

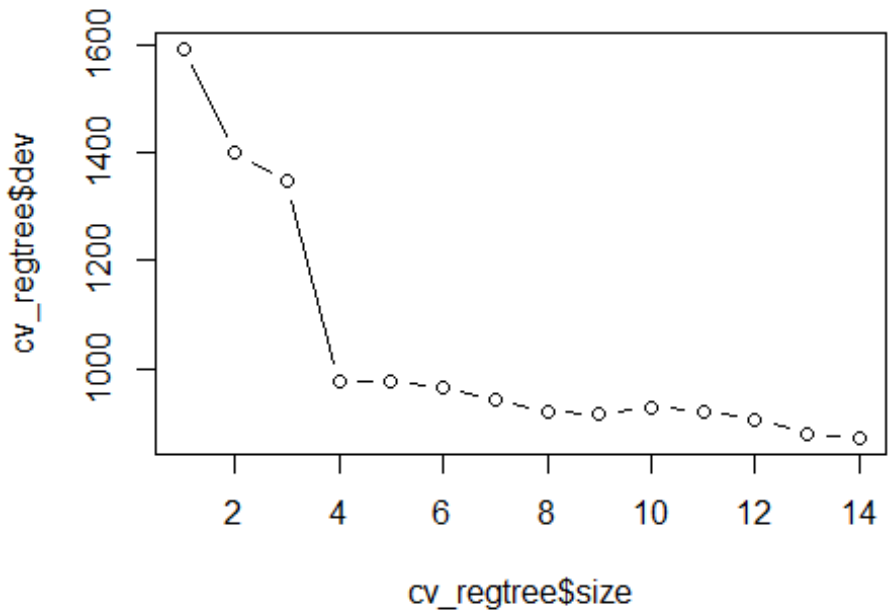
Model building

```r
tree_train <- tree(Damage_Claims~.,training_data)

plot(tree_train)
text(tree_train, pretty = 0)
```




```
summary(tree_train)

##
## Regression tree:
## tree(formula = Damage_Claims ~ ., data = training_data)
## Variables actually used in tree construction:
## [1] "Distance_from_Fire" "Property_Value"     "Humidity"
## Number of terminal nodes:  14
## Residual mean deviance:  1.352 = 549 / 406
## Distribution of residuals:
##     Min.  1st Qu.   Median     Mean  3rd Qu.     Max.
## -4.26400 -0.80600  0.02716  0.00000  0.76590  3.19900

set.seed(12)
cv_regtree <- cv.tree(tree_train)

cv_regtree

## $size
##  [1] 14 13 12 11 10  9  8  7  6  5  4  3  2  1
##
## $dev
##  [1]  871.4151  878.2538  905.4496  921.3159  928.3191  915.3571  921.3180
##  [8]  941.5276  964.9441  976.3285  976.3436 1346.1607 1399.4211 1591.6324
##
## $k
##  [1]      -Inf  15.82732  16.33324  20.42914  26.43445  29.27510  32.80486
##  [8]  36.91487  39.83928  41.93083  46.04175 160.37850 185.23915 334.14506
##
## $method
## [1] "deviance"
##
## attr(,"class")
## [1] "prune"         "tree.sequence"
```

```r
plot(cv_regtree$size, cv_regtree$dev, type = "b" )
```



#Prune the best model with the lowest "dev"

```r
# Prune to size 15 for minimal deviance
pruned_tree_9 <- prune.tree(tree_train, best = 9)
predictions_9 <- predict(pruned_tree_9, testing_data)
```

#Check the model Accuracy

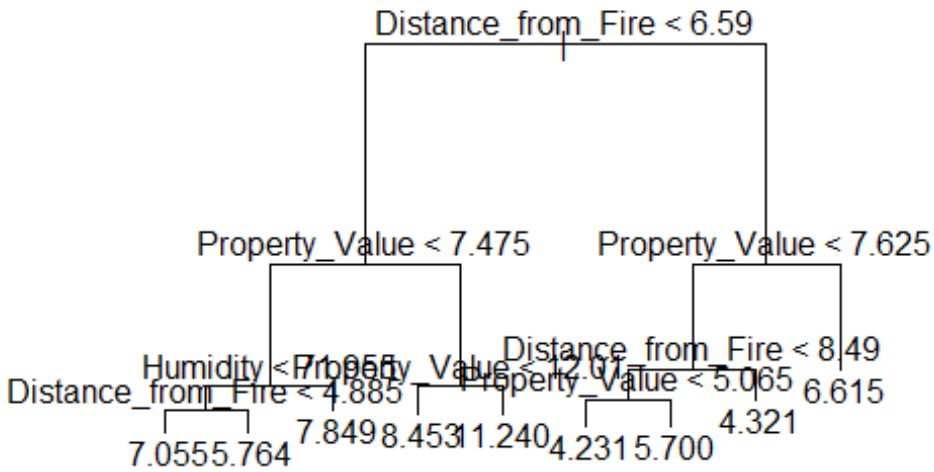
```r
MSE_9 <- mean((testing_data$Damage_Claims - predictions_9)^2)
RMSE_9 <- sqrt(MSE_9)
RMSE_9

## [1] 1.461267
```

Plot the best model Regression Tree

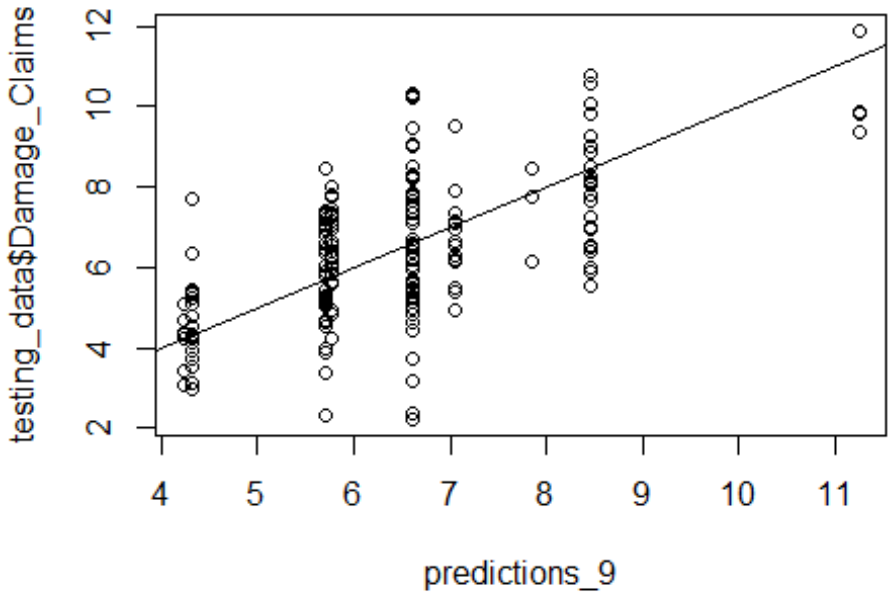
```r
pruned_tree_9

## node), split, n, deviance, yval
##       * denotes terminal node
##
##  1) root 420 1535.000  6.178
##    2) Distance_from_Fire < 6.59 160  530.300  7.315
##      4) Property_Value < 7.475 102  213.200  6.504
##        8) Humidity < 71.955 85  139.200  6.235
##          16) Distance_from_Fire < 4.885 31   33.590  7.055 *
##          17) Distance_from_Fire > 4.885 54   72.790  5.764 *
##        9) Humidity > 71.955 17   37.090  7.849 *
##      5) Property_Value > 7.475 58  131.800  8.742
##       10) Property_Value < 12.01 52   81.490  8.453 *
##       11) Property_Value > 12.01 6    8.417 11.240 *
##    3) Distance_from_Fire > 6.59 260  670.200  5.478
##      6) Property_Value < 7.625 176  359.600  4.936
```

```
##       12) Distance_from_Fire < 8.49 104  205.900  5.361
##         24) Property_Value < 5.065 24   33.210  4.231 *
##         25) Property_Value > 5.065 80  132.900  5.700 *
##       13) Distance_from_Fire > 8.49 72  107.600  4.321 *
##      7) Property_Value > 7.625 84  150.200  6.615 *
```

```
plot(pruned_tree_9)
text(pruned_tree_9, pretty = 0)
```



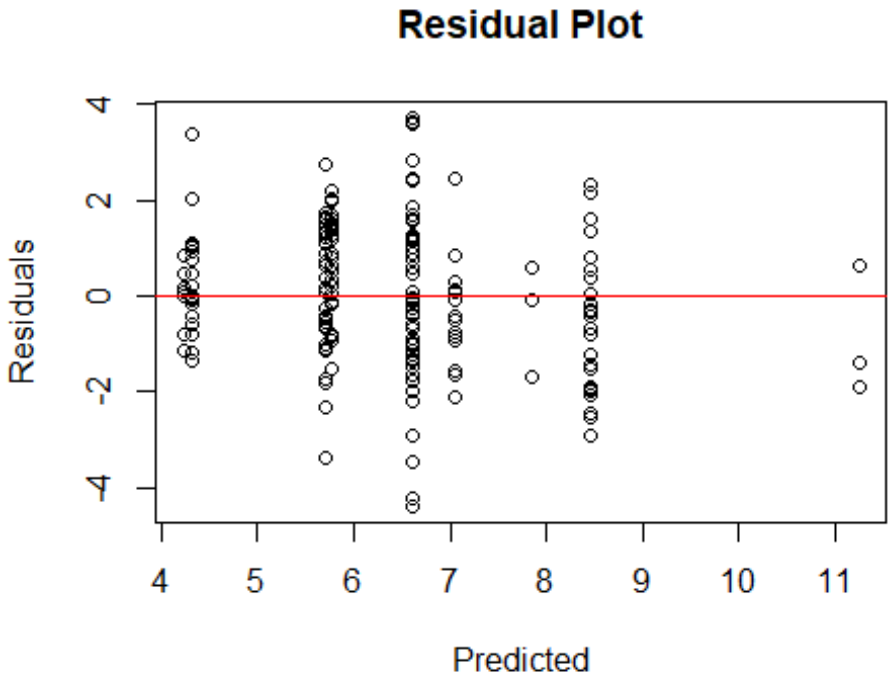

```
plot(predictions_9, testing_data$Damage_Claims)
abline(0, 1)
```



Lets check if the model is biased.

Residuals should be randomly scattered around zero; patterns may suggest areas where the model struggles.

```
residuals <- testing_data$Damage_Claims - predictions_9
plot(predictions_9,
     residuals, xlab = "Predicted", ylab = "Residuals", main = "Residual Plot")
abline(h = 0, col = "red")
```



QUESTION 8

The model is size with 9;

```
MSE <- mean((testing_data$Damage_Claims - predictions_9)^2)
RMSE <- sqrt(MSE_9)
RMSE

## [1] 1.461267
```

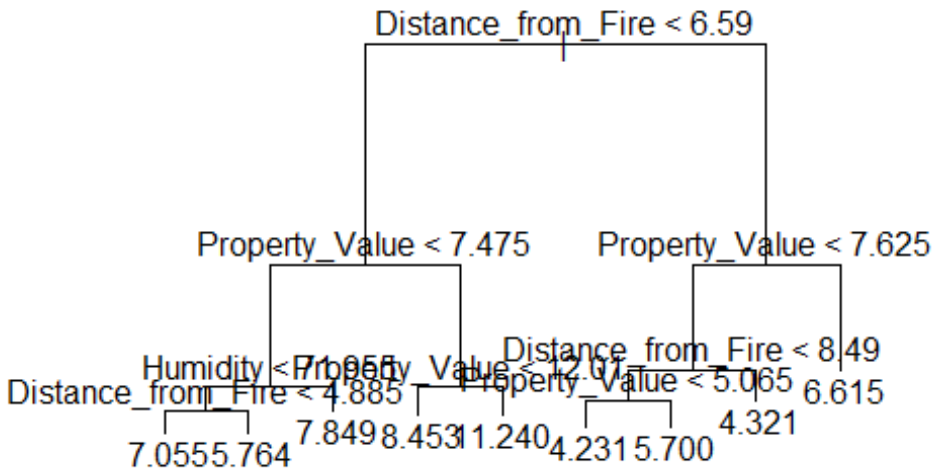
Terninal Nodes:

```
pruned_tree_9

## node), split, n, deviance, yval
##       * denotes terminal node
##
##  1) root 420 1535.000  6.178
##    2) Distance_from_Fire < 6.59 160  530.300  7.315
##      4) Property_Value < 7.475 102  213.200  6.504
##        8) Humidity < 71.955 85  139.200  6.235
##          16) Distance_from_Fire < 4.885 31   33.590  7.055 *
##          17) Distance_from_Fire > 4.885 54   72.790  5.764 *
##        9) Humidity > 71.955 17   37.090  7.849 *
##      5) Property_Value > 7.475 58  131.800  8.742
##        10) Property_Value < 12.01 52   81.490  8.453 *
##        11) Property_Value > 12.01 6    8.417 11.240 *
##    3) Distance_from_Fire > 6.59 260  670.200  5.478
##      6) Property_Value < 7.625 176  359.600  4.936
##        12) Distance_from_Fire < 8.49 104  205.900  5.361
##          24) Property_Value < 5.065 24   33.210  4.231 *
##          25) Property_Value > 5.065 80  132.900  5.700 *
```

```
##        13) Distance_from_Fire > 8.49 72  107.600  4.321 *
##       7) Property_Value > 7.625 84  150.200  6.615 *
```

Interpretting the model with size 9

```
plot(pruned_tree_9)
text(pruned_tree_9, pretty = 0)
```



QUESTION 9

Best model for regression

```
summary(final_model)

##
## Call:
## lm(formula = Damage_Claims ~ Fire_Intensity + Distance_from_Fire +
##     Property_Value + Emergency_Response_Time + Mitigation_Measures +
##     Humidity, data = training_data)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -2.8343 -0.6784  0.0130  0.6447  3.0003
##
## Coefficients:
##                           Estimate Std. Error t value Pr(>|t|)
## (Intercept)              2.374e+00  3.580e-01   6.629 1.06e-10 ***
## Fire_Intensity           6.831e-05  1.319e-05   5.179 3.50e-07 ***
## Distance_from_Fire      -5.072e-01  2.362e-02 -21.473  < 2e-16 ***
## Property_Value           5.114e-01  2.314e-02  22.099  < 2e-16 ***
## Emergency_Response_Time  9.438e-02  1.216e-02   7.759 6.76e-14 ***
## Mitigation_Measures      9.030e-02  2.783e-02   3.245  0.00127 **
## Humidity                 2.928e-02  2.600e-03  11.264  < 2e-16 ***
```

```
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.001 on 413 degrees of freedom
## Multiple R-squared:  0.7304, Adjusted R-squared:  0.7265
## F-statistic: 186.5 on 6 and 413 DF,  p-value: < 2.2e-16
```

Best model for Tree

```
summary(pruned_tree_9)

##
## Regression tree:
## snip.tree(tree = tree_train, nodes = c(10L, 25L, 13L, 7L))
## Variables actually used in tree construction:
## [1] "Distance_from_Fire" "Property_Value"     "Humidity"
## Number of terminal nodes:  9
## Residual mean deviance:  1.599 = 657.3 / 411
## Distribution of residuals:
##      Min.   1st Qu.    Median      Mean   3rd Qu.      Max.
## -4.264000 -0.806400 -0.009098  0.000000  0.847900  3.275000
```

Check the model Performance of both models

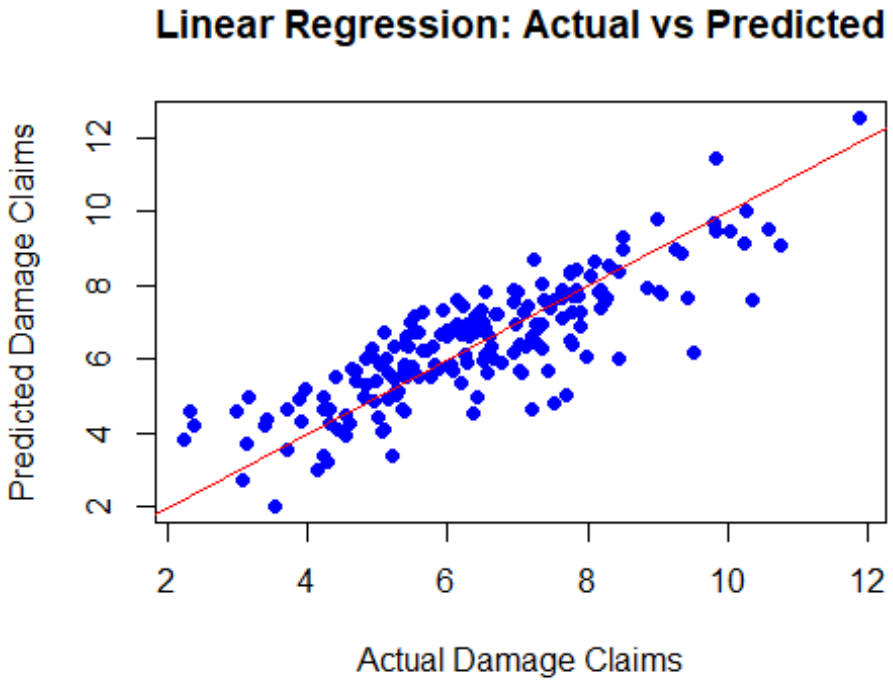
```
LinearModel <- predict(final_model, testing_data)
linearMSE <-  mean((testing_data$Damage_Claims - LinearModel)^2)
linearRMSE <- sqrt(linearMSE)
linearRMSE

## [1] 1.002394
```

Tree RMSE

```
RMSE

## [1] 1.461267
```

Linear Regression Plot

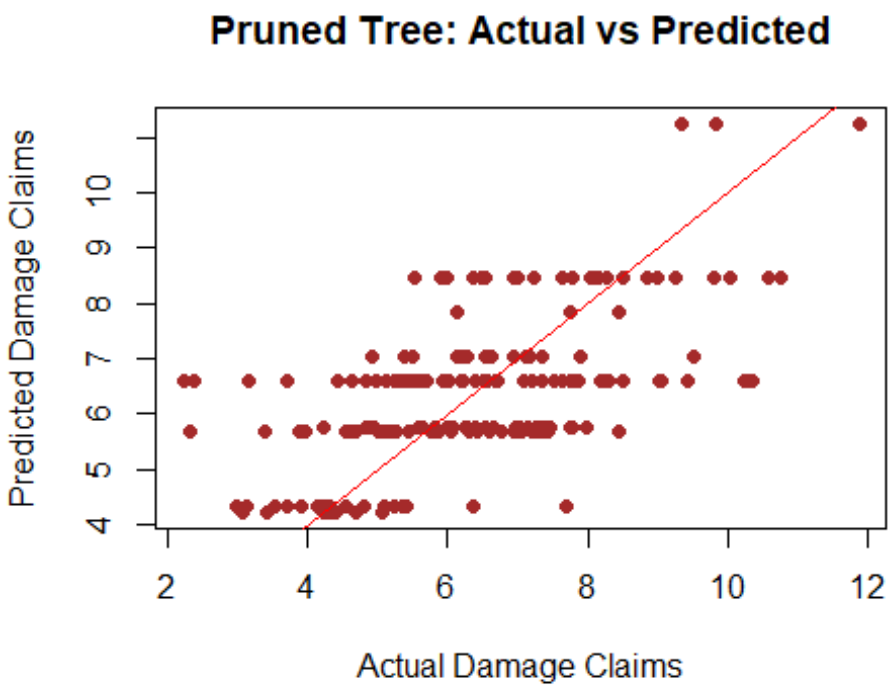
```
plot(testing_data$Damage_Claims, LinearModel, main = "Linear Regression: Actual vs
Predicted",
     xlab = "Actual Damage Claims", ylab = "Predicted Damage Claims", col = "blue"
, pch = 19)
abline(0, 1, col = "red")  # 45-degree line
```



Pruned Tree

Model Plot

```
plot(testing_data$Damage_Claims, predictions_9, main = "Pruned Tree: Actual vs Pre
dicted",
     xlab = "Actual Damage Claims", ylab = "Predicted Damage Claims", col = "brown
", pch = 19)
abline(0, 1, col = "red")  # 45-degree line
```



#QUESTION 10 # Create the new binary target variable for classification # 'NewDamageClaims' is 1 if 'Damage_Claims' is greater than 7.5, otherwise it's 0

```
bushfire$NewDamageClaims <- ifelse(bushfire$Damage_Claims > 7.5, 1, 0)
```

Remove unnecessary variables

```
head(bushfire$NewDamageClaims)

## [1] 0 0 0 1 1 0

str(bushfire)

## 'data.frame':    600 obs. of  14 variables:
##  $ ClaimID                : chr  "DQTA26862" "RDAQ32534" "WTLA41817" "YHZL16317
"  ...
##  $ Fire_Intensity         : num  5738 12248 7317 13479 14226 ...
##  $ Distance_from_Fire     : num  5.57 5.49 5.12 4.89 6.13 ...
##  $ Building_Age           : int  19 27 16 30 21 27 24 24 36 30 ...
##  $ Property_Value         : num  9.76 5.89 3.98 4.91 9.31 ...
##  $ Population_Density     : int  520 433 634 576 644 476 500 591 487 441 ...
##  $ Emergency_Response_Time: int  16 18 17 14 12 10 16 19 15 18 ...
##  $ Mitigation_Measures    : int  3 3 4 2 2 1 5 7 2 3 ...
##  $ Construction_Quality   : Factor w/ 2 levels "Bad","Good": 2 2 2 2 2 2 2 2 2
2 ...
##  $ Insurance_Coverage     : Factor w/ 3 levels "Fully","None",..: 1 2 3 3 2 1 3
1 1 3 ...
##  $ Wind_Speed             : num  24.9 21 25.3 22.3 37.2 31.8 27.8 22.5 32.2 20.
6 ...
##  $ Humidity               : num  35.3 28.2 44.5 53.8 73.8 ...
##  $ Damage_Claims          : num  7.02 7.45 4.3 7.78 9.46 4.54 7.14 7.73 6.03 5.
37 ...
##  $ NewDamageClaims        : num  0 0 0 1 1 0 0 1 0 0 ...
```